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PHYSICAL LAYER SECURITY IN WIRELESS SENSOR NETWORKS

By

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degree of Master of Science in Advanced Control and Systems Engineering**

EXECUTIVE SUMMARY

INTRODUCTION/BACKGROUND

Physical layer security is a new paradigm that complement and enhance the security of wireless communications systems by using the randomness associated to communication channels. Achieving confidentiality in communication networks, traditionally, depends on cryptographic schemes such as public –key (Asymmetric) and private-key (Symmetric) encryption. These techniques are deemed secure base on the premise of limited computational abilities of an eavesdropper. But due to advance in computation capabilities and the dynamic topology and proliferation of modern wireless networks of applicability this cryptographic techniques in securing communication is increasingly questionable and less reliable. Physical layer security concepts scheme vary from cryptographic technique and based on the premise to complement and enhance the security by using the randomness associated to communication channels without resulting to use of computation power at any eavesdropper. This project examines the applicability of this new model to wireless sensor network for prospective implementation in Rolls-Royce gas turbine engines.

AIMS AND OBJECTIVES

The main aim of this dissertation is to investigate the implementation of wireless sensor network with physical layer security in engine monitoring environments.

Objectives are

1. Implement a wireless sensor network system on matrix laboratory software(MATLAB)
2. Examine performance of physical layer security scheme through numerical simulation
3. Establish design constraints set up by physical and hardware platform
4. Analyse performance of sensor deployment strategies via numerical simulation

ACHIEVEMENTS

The following are summary of the dissertation achievements:

1. Review of information theoretic security over wireless channels has been studied and understood. Specifically opportunistic communication protocol
2. Close form characterization of secrecy capacity performance measure of security in information theoretic security has been analysed. This provides an insight for practical design scheme.
3. Review of significates and limitation of both cryptography and information theoretic security in different wireless communications systems scenarios were studied and understood.

CONCLUSIONS / RECOMMENDATIONS

This project concludes that is likely that deployment of any physical layer security protocol will complement and enhance additional layer of security to modern communications systems, because a close form characterization of secrecy capacity which is performance measure of information theoretic security has been analyzed and this provide an intuition in practical design technique such as secure region to place the receiver in an engine model. An investigation to conveniently find the optimal rate pairs on both transmitter and receiver links for different intercept probability constraint to effect the maximum end to end throughput binary search algorithm is recommended for future work.

ABSTRACT

Physical layer security is a new paradigm that complement and enhance the security of wireless communications systems by using the randomness associated to communication channels. Achieving confidentiality in communication networks, traditionally, depends on cryptographic schemes such as public –key (Asymmetric) and private-key (Symmetric) encryption. These techniques are deemed secure base on the premise of limited computational abilities of an eavesdropper. But due to advance in computation capabilities and the dynamic topology and proliferation of modern wireless networks of applicability this cryptographic techniques in securing communication is increasingly questionable and less reliable. Physical layer security concepts scheme vary from cryptographic technique and based on the premise to complement and enhance the security by using the randomness associated to communication channels without resulting to use of computation power at any eavesdropper. This project examines the applicability of this new model to wireless sensor network for prospective implementation in Rolls-Royce gas turbine engines. A close form of secrecy capacity characterization has been analysed and this provide an intuition in practical design technique such as secure region to place the receiver in an engine model and making recommendation of the most suitable approach. Review of significates and limitation of cryptography and information theoretic security in different wireless communications systems scenarios were lastly highlighted. Thus, Implementation of physical layer security will complement and enhance additional layer of security to modern communications systems.

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LIST OF ABBREVIATIONS

WSN (Wireless Sensor Network)

PHY (Physical Layer)

MAC (Medium Access Control)

ACDM (Algebraic Channel decomposing multiplexing)

CSI (Channel state information)

LDPC (Low density Parity Check Codes)

AWGN (Additive White Gaussian noise)

SNR (Signal to Noise Ration)

ARQ (Automatic repeat request)

DOS (Denial of Service)

Chapter 1 - Introduction

1.1. Background and Motivation

Wireless Sensor network are cost-effective alternative with numerous applications. This applications ranges from climate monitoring , light, humidity detection, temperature of the surrounding environment, airport or stadium chemical, atmospheric pollution monitoring, the abnormal family environment, monitoring the structural integrity of buildings, biological threat detection and prediction. Sensor network is the reason for the implementation in complex large-scale network environment, real-time data acquisition and processing. WSN is normally implemented in a way, no maintenance, non-controlled environment, besides the wide disclosure of information, denial of service, replay attacks, message modifications etc. Wireless sensor network (WSN) point of intersection are vulnerable to physical tricking of the attacker. Hence, need to provide sufficient measures to counter these attacks from occurring.

Primarily, Security schemes are to counter these threats from happing. These schemes can be based on symmetric key encryption schemes and message authentication codes and the public key cryptography. However, these scheme are prone key frame up and conception, practicability of this scheme (cryptographic) in hardware, sophisticated measures such as spread-spectrum to deal with jamming, feasibility of the public key cryptography. [1]

1.1.1. MOTIVATION

Wireless senor network at present can easily be attacker as result of high vulnerability of wireless channel besides the traditional security schemes. Wireless channels are prone to Channel jamming, authentication challenge and free accessibility of the channel. A malicious eavesdropper can jam physical communication channels and block authorized users from utilizing the network. Inadequate authentication scheme can also enhance malicious eavesdropper to have unauthorized entering to the network resources by circumventing all security protocol. Malicious attacker can also utilize

openness of wireless channel to eavesdrop on communication without adopting any sophisticated technology equipment. Generally, legitimate users in a network could also be regarded as potential eavesdroppers. A technique to prevent this threat from happening have been developed using a layered approach. This approach is basically used to easy the design of communication protocol with less significances to security. [3]. Cryptographic protocols are designed and realized with premise that physical layer vulnerability counter mechanism has been established. In contrast to this assumption, there are feasible research that proved that PHY security scheme will complement and enhance security of wireless communication system [23]

1.2. Problem Definition

Physical layer security basic can be illustrated in figure 2.4 where exchange of information between transmitter and legitimate receiver (Terminals T1 and T2) is been intercepted by an eavesdropper source (Terminal T3). The exchange of information between transmitter and legitimate receiver (Terminal T1 and T2) is called main channel where exchange of information between Transmitter (Terminal T1) and eavesdropper (Terminal T3) is referred to eavesdropper's channel [3]

1.3. Aims and Objectives

The purpose of this dissertation to investigate implementation of wireless sensor network with physical layer security in engine monitoring environment. Objectives are;

1. Implement a wireless sensor network system on matrix laboratory software(MATLAB)
2. Examine performance of physical layer security scheme through numerical simulation
3. Establish design constraints set up by physical and hardware platform
4. Analyse performance of sensor deployment strategies via numerical simulation

1.4. Project Management

Human Recourses required for this project work are

Resources	Availability	Notes
Student	Yes	Responsible for carrying out the task
Supervisor	Yes	Responsible for guidance

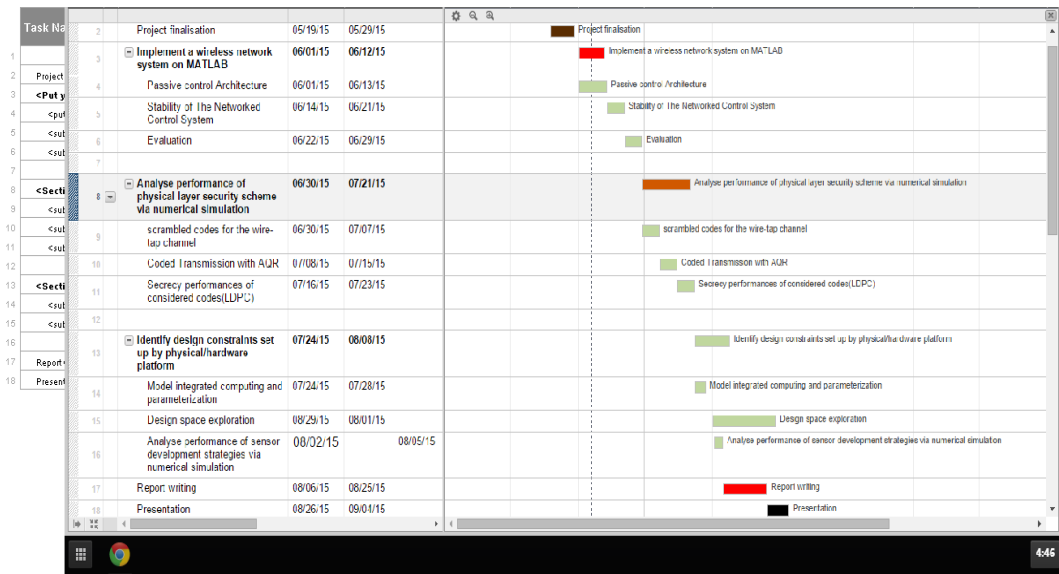
Facilities required for this task

Resources	Availability	Notes
Computer	Yes	-
MSWord/MATLAB	Yes	-

Risk and Cost in carrying out this task

Cost	Availability	Risk	Motivators
Individual Study time	840 hours	High risk	Adequate preparation to meet deadline
Supervisor Guidance	30 munities of meeting in every two week	Low risk	Preparation for the meeting
Report writing limitation	70 pages	Average risk	Enough to start writing the report and proper rehearsal

1.5. PROJECT STRUCTURE



1.5. Chapters overview

Chapter 2: This chapter gives a brief introduction to physical layer security; outline the state of art by an introduction to wiretap channel models, characterization and scaling of Secrecy Capacity as well as reviews different types of techniques for selecting secrecy capacity in wireless sensor network.

Chapter3: The chapter outlined experimental set up and methodology use in information theoretic security for measurement of secrecy capacity.

Chapter 4: This chapter present a close-form characterization of secrecy capacity of a simple wiretap model and analysis its importance. The result obtained from the full Investigation of the model would provide an insight for practical design scheme to secure and provide reliable wireless communication.

Chapter 5: This chapter gives basic difference between classical cryptograph and information-theoretic security and point out what these difference are and there effects on the choice of technology in a wireless scenarios.

Chapter 6: This chapter provides summary of the dissertation and explained recommendation for future work.

Chapter 2 - LITERATURE REVIEW

2.1. INTRODUCTION

The background of physical layer security, the state of art and guide to the proceeding chapters is provided in this chapter. Brief introductory to wiretap channel model, characterization and scaling of Secrecy Capacity, different types of secrecy capacity techniques and methodologies were also reviewed.

2.1.1. Concept and constraints of Cryptography

Cryptograph is a security scheme deployed to counter an attacker from intercepting communications. Techniques such as data integrity, data confidentiality, authentication and non-repudiation goal is to block the adversaries. Cryptography is also express as the conversations of data into code word that can be unscramble and transmit across a public or private network. There two types of Cryptography schemes namely public key (Asymmetric) and private key (symmetric-key). Different keys employ for encryption and decryption by public key while schemes use one key for encryption and decryption by private key as represented in figure 2.1 and 2.2[2]

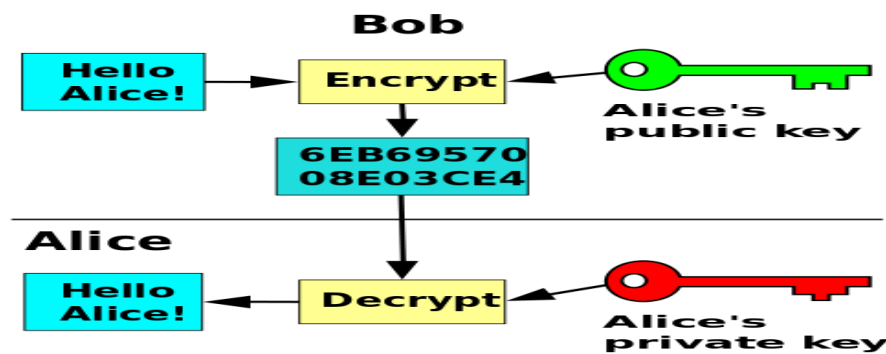


Figure 2.1. Public-key cryptography [2]

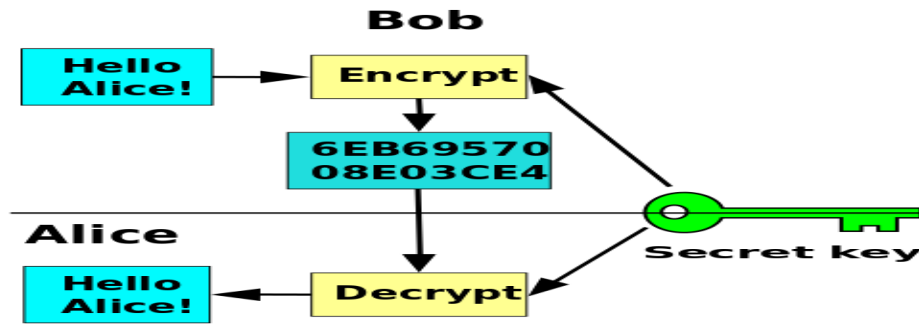


Figure 2.2 Private-key cryptography [2]

Modern cryptography also addresses the problems such as data signature, message authentication and data integrity etc. Specifically, the focus of this dissertation is on description of secret key and public key encryption schemes. The goal is to point out important features of this scheme and gives a justifiable comparison with information theoretic techniques.

In order to ensure the confidentiality of message transmitted by transmitter (Alice), codes are suited on keys, which are secret sequence of bits only known to transmitter (Alice) or legitimate receiver (Bob). Eavesdropper (Eve) objectives is to decipher the Transmitter (Alice) and legitimate receiver (Bob) communication (codes) to wiretap information. That is to recapture messages from the code word without having knowledge of the content of the key.

Securities of encryptions techniques are traditionally determine using sophisticated computations and restrictions of computing capabilities of eavesdropper. Notable, with today's technology, computational security ensures that the amount of computing time or memory required to break a code is absurd. This concept of security is extensively used in current cryptographic protocols. However, despite being satisfactory in several scenarios, it fails to guarantee security in the long term. Specifically, several codes used in few years back are easily breakable via computer (PC). Therefore, encoding algorithms will required frequently renewal to tackle rising power of computers [3]

2.3 Physical layer security

Various layers considered in a classical wireless communication protocol and their unique purpose is illustrated in Figure 2.3. For example, a channel coding is implemented at the physical layer to ensure free communication of all layer without an error. However, the design of modern communication protocol fall short of this type of layer approach and employ cross-layer.

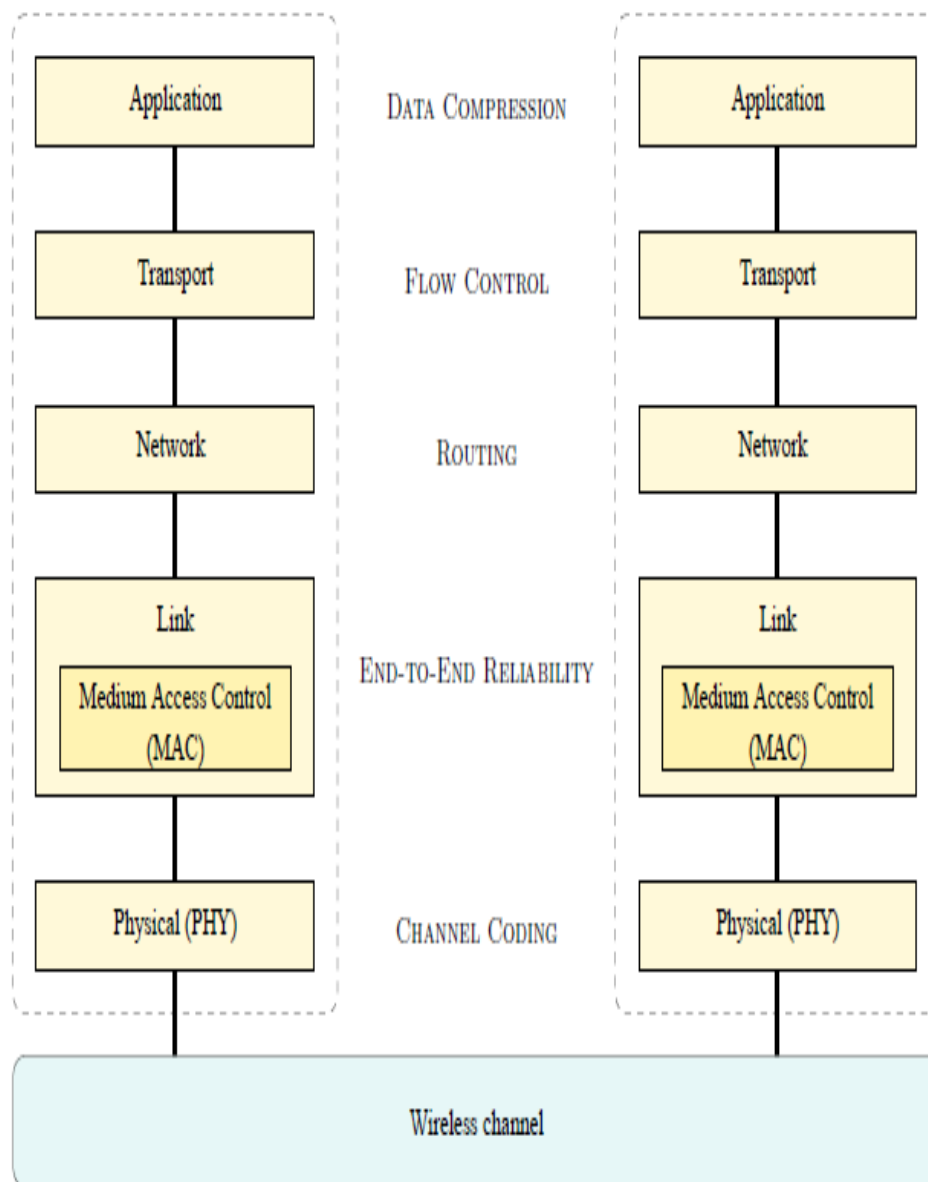


Figure 2.3 Layered protocol architecture [3]

Physical layer security schemes such as modulation techniques, authentication mechanisms and message encryption are all implemented to complement and enhance security of wireless communication systems.[3]

Problem Statement

Figure 4 illustrated the basic principle of physical layer security employing three wireless terminal network architecture. The exchange of information between two legitimate source Alice and Bob (Terminal T_1 and T_2) is being intercepted by illegitimate source Eve (Terminal T_3). The exchange of information between the two legitimate source Alice and Bob (Terminal T_1 and T_2) are referred as main channel and the interception of information from the transmitter Alice (Terminal T_1) by an illegitimate source Eve (terminal 3) is refer to as eavesdropper channel. Radiofrequency signals are discovered when Bob (terminal T_2) and Eve (T_3 Terminal) communication signal are scattered and there is variation on the output of the main channel and eavesdropper's channel.

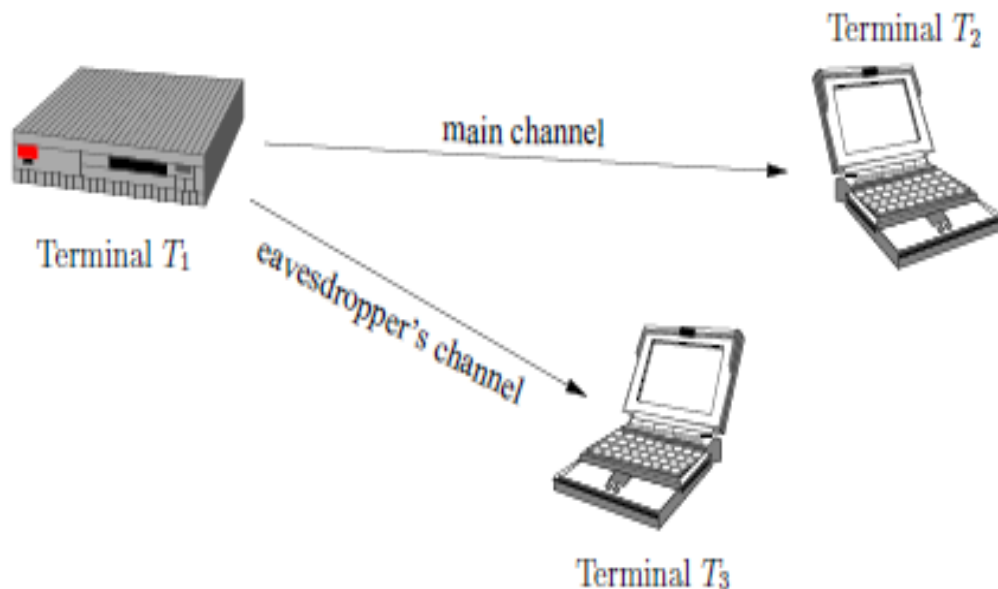


Figure 2.4 Illustrations of eavesdropping scenarios in wireless network [3]

These variations are induced by physical phenomena such as fading and path-loss. Fading occurs due to multi-path propagation of radiofrequency waves and path-loss occurs due to attenuation of wave amplitude with distance. Signal is detected when transmission distance over the main channel is less than that of the eavesdropper's channel. However, most Morden technology is deployed with premise that an equivalent signal is detected by eavesdropper as that of legitimate receiver. In contrast to these premise, physical layer security objective is utilizes these variations to complement and enhance security of communication systems.

The three-node communication problem in Figure 2.4 can be studied from an information-theoretic perspective as detail in chapter 3, and the security profited from the PHY layer can enhanced and complement the security of wireless network.. This research is carried out exclusively from an information-theoretic and coding techniques though identified that physical layer security goes beyond these considerations .Specifically, extensive schemes that tends to modify the PHY layer to marred the possible eavesdropper such as coded –division multiple access signalling that permit a noise-line appearance, and beamforming with smart antennae's that bar eavesdropper far away from receiver intercept exchange of information-[3]

2.4 Wiretap Channel model

Perfect secrecy concept was first introduced by Shannon which laid the theoretical basis for this approach; ‘perfect secrecy’ can be obtain [4] when eavesdropper's is not privy to the exchange of information then it is difficulty to intercept and obtain the original message contain in transmitted signal.

Wyner refined the concept of perfect secrecy by introducing a wire-tap channel. The source node broadcast a signal to a destination node via a discrete memoryless channel and unauthorized node ‘wire-tapper’ eavesdrops the signal via a different type of memoryless channel [5]. Wiretap proved that confidential information can be achieve if the eavesdropper channel is more rowdy than the main channel and secrecy capacity is realised provided rate of code chosen is smaller.

Csisz'ar and K in 1978 proved that a non-zero secrecy capacity can be achieve when the main channel is less rowdy than the wiretap channel and proposed broadcast channel illustrated in figure 2.5 and [6].

The two legitimate users Alice and bob communicate via a discrete broadcast channel define by a discrete input letter X and two discrete output letter Y and Z , and probability transition function $P_{YZ|X}(yz|x)$.

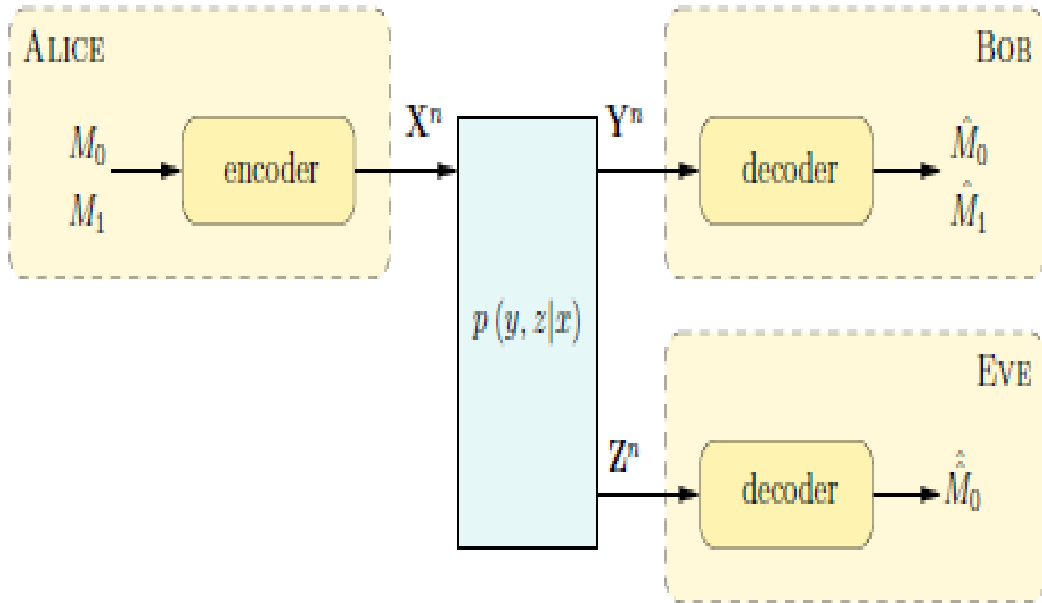


Figure 2.5. Broadcast wiretap channel [6]

The number of occurrence of n symbols (probability of sequence) is indicated as

$$P(y^n, z^n | x^n) = \prod_{i=1}^n P_{YZ|X}(y_i, z_i | x_i)$$

With the presumption that legitimate transmitter (Alice) intends to transmit a message M_0 both legitimate receiver (Bob) and illegitimate receiver (Eve) by conceal a secret letter M_1 to legitimate receiver (Bob) only using $2^{nR_0}, 2^{nR_1}, n$ for transmission, $f_n: M_0 \times M_1 \rightarrow X^n$ aligning each message pair $(M_0, M_1) \in M_0 \times M_1$ to a codeword $x^n \in X^n$, Two decoding functions $g_n: y^n \rightarrow M_0 \times M_1$ and $h_n: z^n \rightarrow M_0$ which map

an observation y^n to a message pair $(\hat{M}_0, \hat{M}_1)$ and an observation z^n to a message $\hat{m}_0$.

The private message M_1 depends on equivocation rate $\frac{1}{n}H(M_1|Z^n)$ and a rate tuple $(R_{T_0}, R_{T_1}, R_{T_e})$ and is feasible for a broadcast channel with private messages if and only if, for any $\epsilon > 0$ there exists a code such that

$$P[g_n(Y^n) \neq (M_0 \times M_1) \text{ or } Z^n \neq M_0] < \epsilon \quad (\text{Reliability condition})$$

$$\frac{1}{n}H(M_1|Z^n) \geq R_e - \epsilon \quad (\text{Secrecy condition})$$

Obviously is not a priori whether the aforementioned condition can be simultaneously satisfied. Actually, the reliability condition calls for an increase redundancy in the coding scheme, while the secrecy condition tends to limit this redundancy. Trade-off between reliability and secrecy can be further explain in detail in the theorem below [3]

Theorem 2.1 feasible rate tuples set (R_0, R_1, R_e) is indicate as

$$\mathcal{C} = \bigcup_{U \rightarrow V \rightarrow YZ} \left\{ \begin{array}{l} 0 \leq R_e \leq R_1 \\ R_e \leq I(V; Y|U) + \min(I(U; Y), I(U; Z)) \\ 0 \leq R_0 \leq \min(I(U; Y), I(U; Z)) \end{array} \right\}$$

It is appropriate as well to define a scalar metric characterizing the inherent security that a channel can provide least upper bound of all rates R_1 for tuple $(0, R_0, R_1)$ (secrecy capacity of a broadcast channel private messages). This metric provides a counterpart to the usual channel capacity, which considers only reliable communications without secrecy constraints. Based on Theorem 1, the following result can be proven. ([6], theorem 1)

Corollary 2.1 Secrecy capacity of a broadcast channel with Private messages is indicated as

$$C_s = \max_{V-X-YZ} [I(V; Y) - I(V; Z)] \quad 2.1$$

Generally, corollary 1 provides a formula that facilitates computation of all secrecy capacity of discrete memoryless. It can also be proven to facilitate computation for

continuous memoryless channels. Nevertheless, equation (2.1) involves maximization over random variable satisfying a Markov chain condition, which provides little insight in practical scenarios. As indicated in equation 2.1 Secrecy capacity realization depends on channel transition probability through marginal probability $P_{Y|X}(y|x)$ and $P_{Z|X}(z|x)$. If main channel and eavesdropper channel are noiseless, then the secrecy capacity of a broadcast channel is zero and security is cannot be guarantee base on cryptography scheme. ([6], corollary 2)

Wyner proposed Gaussian wiretap channel in 1978 and proved that secrecy capacity is realisable when the difference of channel are known (main channels and wire-tap channels) [7]

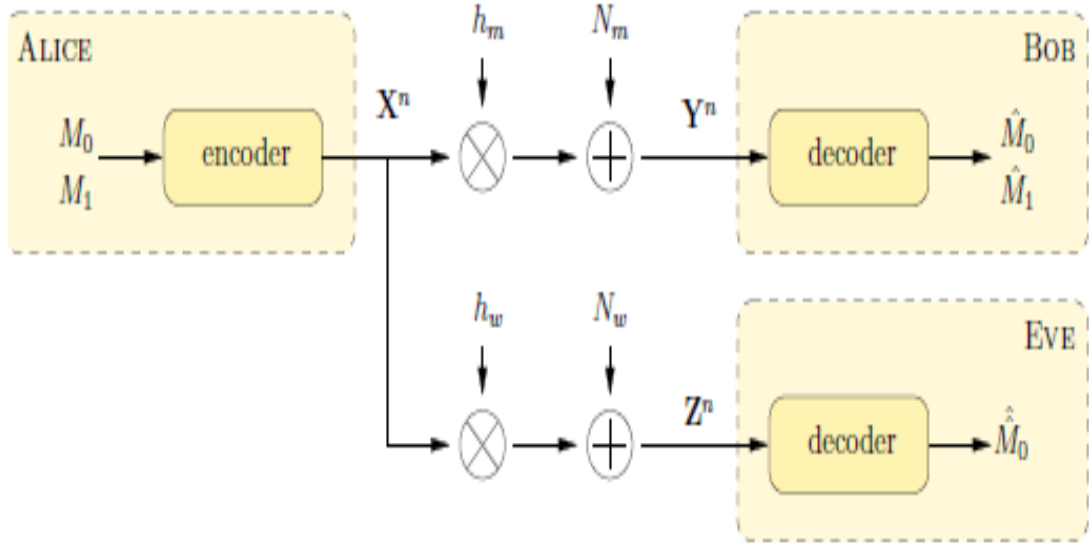


Fig 2.6. Gaussian wiretap channel [7]

The symbol h_m and h_w denotes channel gain, N_m and N_w Gaussian noises, σ_m^2 and σ_w^2 variance. Both main channel and eavesdropper are assumed to be additive white Gaussian and code words transmitted are subject to average power constraints.

$$\frac{1}{n} = \sum_{i=1}^n E\{|X(i)|^2\} \leq P$$

2.5 Pertinence of wiretap channel model

Hypothesis are associated to wiretap channel model are as follows;

1. Channel State information (CSI): The manipulation of an eavesdropper is safeguarded if the wiretap code used for transmission is correctly tailored to the channel.
2. Authentication. The wiretap channel model assumes main channel is authenticated. But authentication mechanisms is realisable at upper layers of the protocol stack. Significantly, full authentication can be achieved [9] if a short secret key is available. Particularly, the size required for authentication scales as the logarithm of the message size.
3. Passive eavesdropping. Eavesdropper's movement is restricted in this technique. Thus, extra mechanism is required to cope with jamming[3]
4. Availability of random generator: wiretap encoders depends on perfect random generators because their stochastic encoder, but in practice strong pseudo-random generators may be employed for initialization mechanisms.[3]
5. Weak secrecy capacity. Security is measure in term of equivocation rate $\frac{1}{n}H(M_1|Z^n)$ [10]

2.6 Pertinence of Wiretap Channel Model

The high security vulnerability of wireless medium has led to investigation of wiretap channel by many researchers. Barros and Rodrigues proposed and show that information-theoretic security will be enhance and complemented irrespective of if eavesdropper(Eve) has a good average signal-to-noise ratio than legitimate receiver(Bob).[11]. Liang et al [12], Li et al[13] and Gopala et al. [14] presented secrecy capacity of ergodic fading channels, multiple-input multiple output wiretap channel by[15] and [16], multi-user wiretap channels by[17,18,19,20,21] and wiretap channels with distortion measure by[22]

Chapter 3- EXPERIMENTAL SETUP AND METHODOLOGY

This chapter gives general review of the design of a practical communication scheme providing foundation for information-theoretic security for wireless channels. Limitation of quasi-static fading channels and secure communication protocol is reviewed.

3.1 Information –theoretic Security across wireless Channels

3.1.1 Framework of Wireless wiretap channel

Wyner's presented three nodes in his paper as illustrated in figure3.1. Two authorized user Alice and Bob exchange communication. The transmitter (Alice) encode (Conceal) message in random variable W^k into a code word in *Random Variable* X^n for transmission via the channel. Bob legitimate user (main channel) noticed the output as $Y_m(j) = H_m(j)X(j) + Z_m(j)$ where the term

$H_m(j)$ Denoting circularly symmetric Gaussian random variable with zero-mean and unit-variance denoting the main channel fading coefficient and $Z_m(j)$ denoting zero-mean circularly symmetric complex Gaussian noise random variable.

An unauthorized user Eve (eavesdropper's channel) wiretapping on Alice's communication and noticing the output secretly and separately as

$Y_w(j) = H_w(j)X(j) + Z_w(j)$ Where the term $H_w(j)$ denoting circularly symmetric Gaussian random variable with zero-mean and unit-variance denoting the main channel fading coefficient and $Z_w(j)$ denoting zero-mean circularly symmetric complex Gaussian noise random variable.

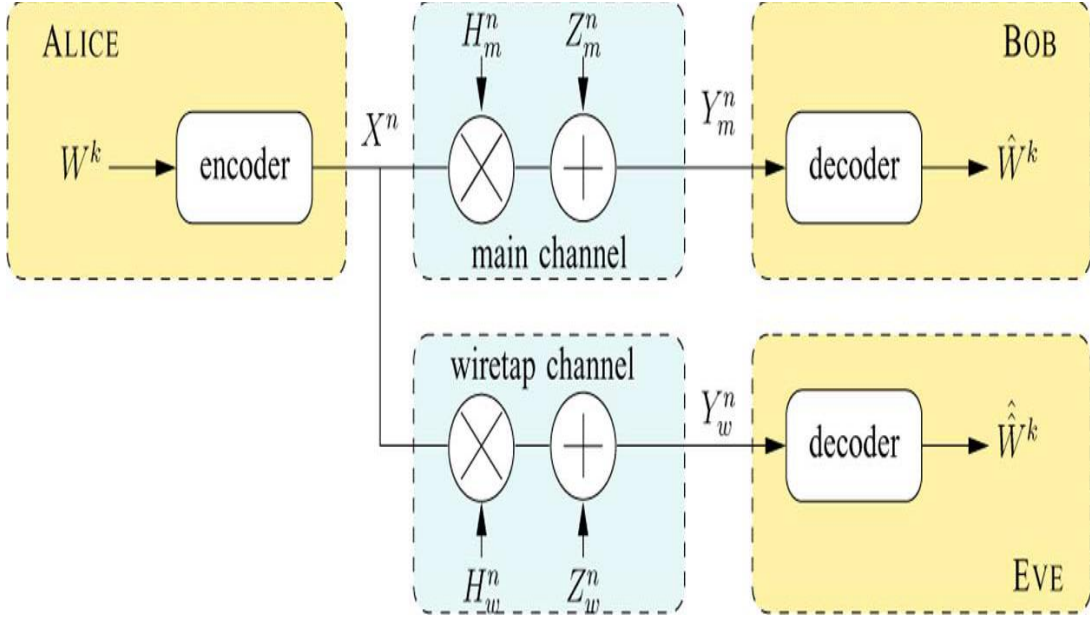


Figure 3.1 Wireless Wiretap channel Model [23].

This wireless wiretap channel model framework presumed that channel input, fading coefficient and channel noises are all independent and quasi-static fading channel. The code word sent by transmitter (Alice) are subjected to average power constraint as

$$\frac{1}{n} = \sum_{j=1}^n E\{|X(j)|^2\} \leq P$$

A (near) instantaneous signal-to-noise ratio at both legitimate (Bob) and eavesdropper (Eve) is denoted as

$$\Gamma_{m(j)} = P|H_m(j)|^2/N_m = P/N_m = \Gamma_m$$

$$\Gamma_{w(j)} = P|H_w(j)|^2/N_w = P|H_w|^2/N_w = \Gamma_w$$

And mean values as

$$\bar{\Gamma}_{m(j)} = P|H_m(j)|^2/N_m = P|H_m|^2/N_m = \bar{\Gamma}_m$$

$$\bar{\Gamma}_{w(j)} = P|H_w(j)|^2/N_w = P|H_w|^2/N_w = \bar{\Gamma}_w$$

Based on the premise of fading coefficient is zero-mean and complex Gaussian random variable the equation below are deduced

$$P(Y_m) = \frac{1}{\bar{Y}_m} \exp\left(-\frac{Y_m}{\bar{Y}_m}\right), \quad Y_m > 0 \quad (3.1)$$

$$P(Y_w) = \frac{1}{\bar{Y}_w} \exp\left(-\frac{Y_w}{\bar{Y}_w}\right), \quad Y_w > 0 \quad (3.2)$$

And secrecy capacity is presented as

$$C_s = C_m - C_w \quad (3.3)$$

Where C_m and C_w denote the secrecy capacity of the legitimate (main) channel and unauthorized (eavesdropper) channel

$$C_m = \frac{1}{2} \log\left(1 + \frac{P}{N_m}\right) \text{ and } C_w = \frac{1}{2} \log\left(1 + \frac{P}{N_w}\right)$$

3.2 SIGNIFICANTS OF FADING TO SECURE COMMUNICATIONS

The importance of fading to complement and enhance secure communication of wireless networks is reviewed. Average secrecy capacity and probability of secrecy capacity are used as a measuring metric to determine the extent of security (Secrecy capacity). [11] Secrecy capacity for one realization of quasi-static fading wiretap channel is presented as

$$C_s(Y_m, Y_w) = \begin{cases} \log(1 + Y_m) - \log(1 + Y_w) & \text{if } Y_m > Y_w \\ 0 & \text{if } Y_m \leq Y_w \end{cases} \quad (3.4)$$

3.2.1 Average Secrecy Capacity

Provided that absolute channel state information about the eavesdropper is at Alice disposal then the coding technique can fit into each attainment of the fading coefficients. Base on this premise, fundamentally, each average of protected communication rate beneath the average secrecy capacity of the channel can be realized [23]

Note 3.1, this can be easily computable numerical as shown in [23, appendix c, Lemma 2] it can be shown that

$$C_s(Y_m, Y_w) = \int_0^\infty \int_0^\infty C_s(Y_m, Y_w) P(Y_m) P(Y_w) dY_m dY_w$$

$$\bar{C}_s = F(\bar{Y}_m) - F\left(\frac{\bar{Y}_m \bar{Y}_w}{\bar{Y}_m + \bar{Y}_w}\right) \quad (3.5)$$

$$\text{Where } F(x) = \int_0^\infty \log_2(1+u) \frac{1}{x} e^{-y/x} du$$

$$= e^{1/x} E1(x^{-1}) \frac{1}{\log 2} \quad (3.6)$$

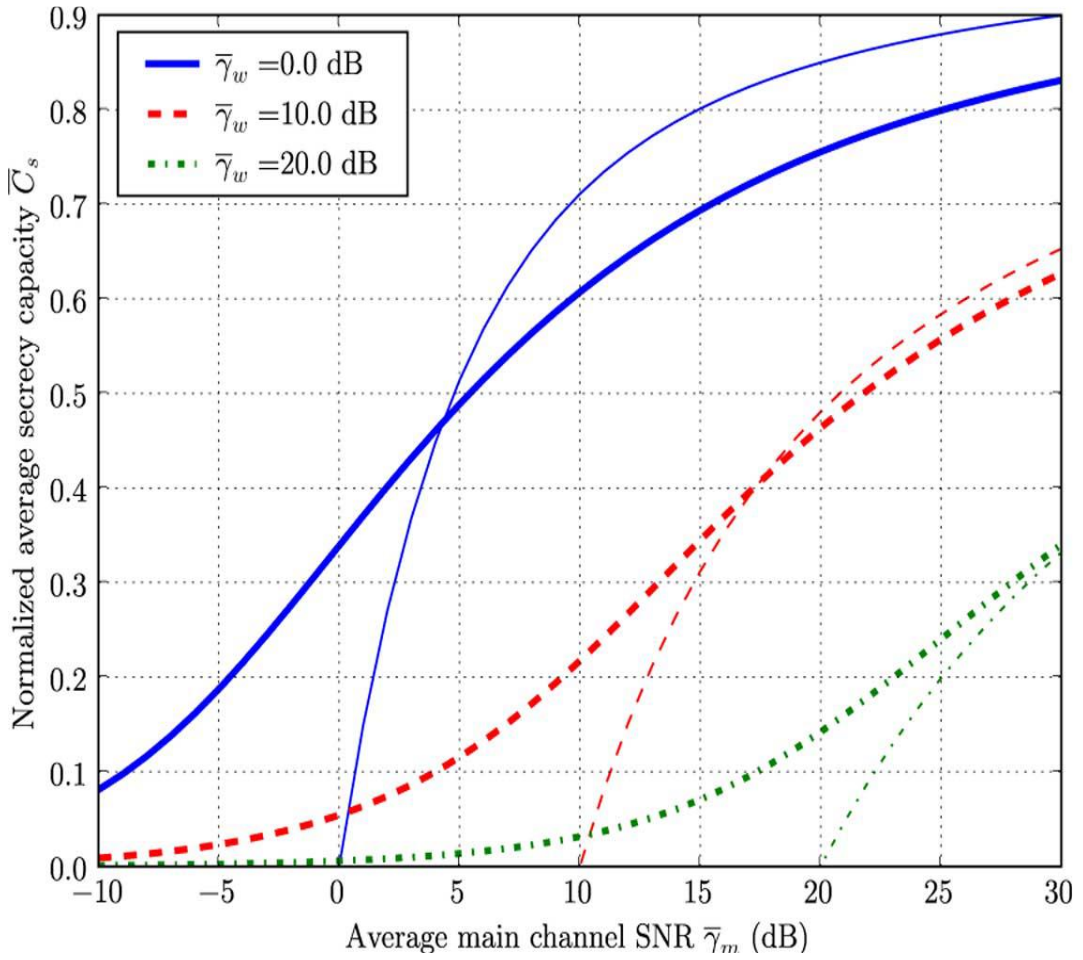


Figure 3.2 Average Secrecy capacities illustration [23]

The slim graph denotes the normalized average secrecy rate of a Rayleigh-fading channel and darker lines denotes Gaussian wiretap channel's secrecy capacity (SC). Normalization is carried out in relation to the capacity of an AWGN channel with Signal

to noise ratio equal to normalized average secrecy capacity .Two conclusion are deduce, Average secrecy capacity rate of fading channel is above or nearby to secrecy capacity of the Gaussian channel while average rate of fading channel is nonzero irrespective of level of signal to noise ration of main channel. [3]

3.2.3 Outage Probability metrics

Outage Probability metric is used for characterization of secrecy capacity of a quasi-static Rayleigh-fading channel

Theorem 3.1:

$$P[C_s > \Gamma] = P_0(\Gamma) = \frac{\bar{Y}_m}{\bar{Y}_m + \bar{Y}_w^{2\Gamma}} \exp\left(-\frac{2^\Gamma - 1}{\bar{Y}_m}\right)$$

Proof:

$$\begin{aligned} P[C_s > \Gamma] &= p\left[\log \frac{1+\Gamma_m}{1+\Gamma_w} > \Gamma\right] \\ &= p[\Gamma_m > 2^\Gamma(1 + \Gamma_w) - 1] \\ &= \int_0^\infty P(Y_w) \left(\int_{2^\Gamma(1+\Gamma_w)-1}^\infty P(Y_m) dY_m \right) dY_w \end{aligned}$$

The last equality indicates that

$$P(Y_m, Y_w) = P(Y_m) P(Y_w)$$

Equation (3.1) and (3.2) gives probability of strictly positive secrecy capacity

$$P[C_s > 0] = \frac{\bar{Y}_m}{\bar{Y}_m + \bar{Y}_w}$$

Outage probability metrics can also be employ when user location is of interest. The probability of strictly positive secrecy capacity is present as

$$P[C_s > 0] = \frac{1}{1 + (d_m/d_w)^\alpha}$$

Terms in the expression are define as:

d_m Represent the distance between transmitter (Alice) and legitimate receiver (Bob)'s distance

d_w Represent the distance between transmitter (Alice) and eavesdropper (Eve)

α Represent path-loss exponent

Note 3.2

From this positive secrecy capacity equation it follow that

Firstly, when the average signal-to noise ration of main channel ($\bar{\gamma}_m$) is greater than or equal to that of eavesdropper or distance between transmitter and legitimate receiver (d_m) is shorter than or equal than that of distance between transmitter and eavesdropper at that point positive secrecy capacity is equivalent to one or secrecy capacity is equal to zero. Secondly , when the average signal-to noise ration of eavesdropper's channel ($\bar{\gamma}_w$) is less than or equal to that of main channel or distance between transmitter and eavesdropper (d_w) is less than or equal that of distance between transmitter and legitimate receiver at that point positive secrecy capacity is equivalent to zero or secrecy capacity is equal to 1.

Base on this insight, high security is realized when the eavesdropper (Eve) is far away from the transmitter (Alice) than the legitimate receiver (Bob)

It also indicated that Nonzero secrecy capacity occurs when the signal to noise ration of main channel is less or equal to that of eavesdropper's channel or distance between transmitter (Alice) and legitimate receiver (Bob) is greater than or equal to that of distance between transmitter and Eavesdropper.

In view of this, outage probability metric could be indicated as the probability when the instantaneous secrecy capacity is below a targeted secrecy rate, presented as

$$P_{out}(R_s) = P[C_s < R_s]$$

Importance of the outage probability metrics in measuring the secrecy capacity of communication systems can be in two way.

Firstly, it provide a fragment of fading attainment for protected rate bits and channel utilization in wireless communication channel. Secondly, it complement and enhance security even if the transmitter (Alice) and legitimate receiver (Bob) is not privy to channel state information of the eavesdropper (Eve).

Theorem 3.2 Outage Probability for a target secrecy rate is presented as

$$P_{out}(R_s) = P[C_s > \Gamma] = 1 - \frac{\bar{\gamma}_m}{\bar{\gamma}_m + \bar{\gamma}_w^{2R_s}} \exp\left(-\frac{2^{R_s}-1}{\bar{\gamma}_m}\right) \quad (3.9)$$

Outage probability for high values of target secrecy rate is expressed by this equation.

If secrecy rate tends to zero then outage probability is presented as

$P_{out} \rightarrow \frac{\bar{\gamma}_m}{\bar{\gamma}_m + \bar{\gamma}_w}$ and if the secrecy rate tends to infinity then the outage probability tends to 1. Notably, when outage probability tends to 1, at this point exchange of confidential message between the transmitter and legitimate receiver is unrealizable as illustrated in figure 3.3 below.

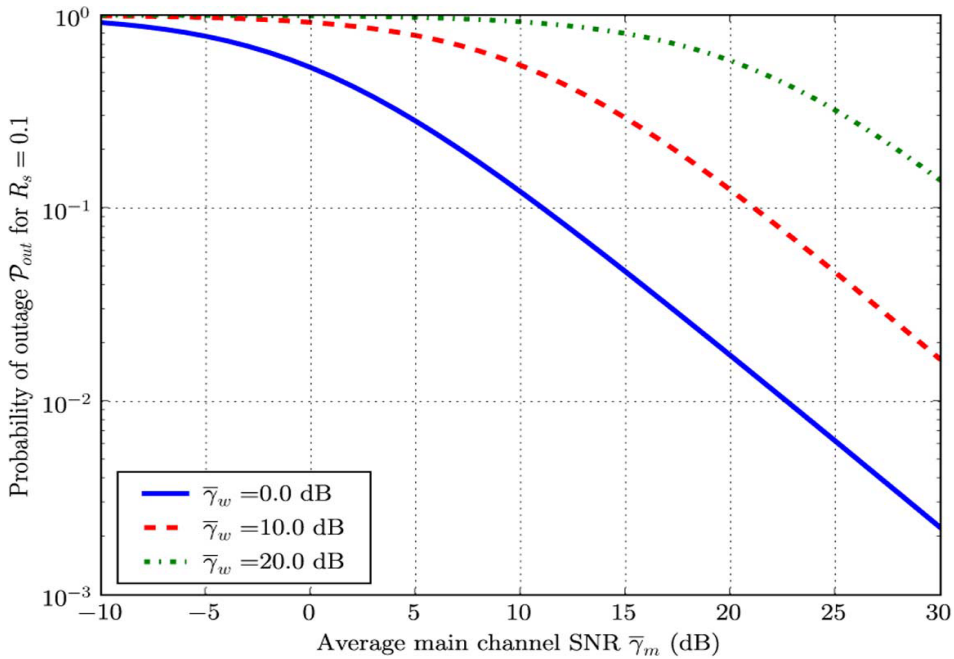


Figure 3.3 Outage Probability versus signal to noise ratio [23]

The plot in figure 3.3 indicates outage probability against signal to noise ratio of main channel and specify values of signal to noise ratio of eavesdropper's channel for a normalize target secrecy rate of 0.1[23]

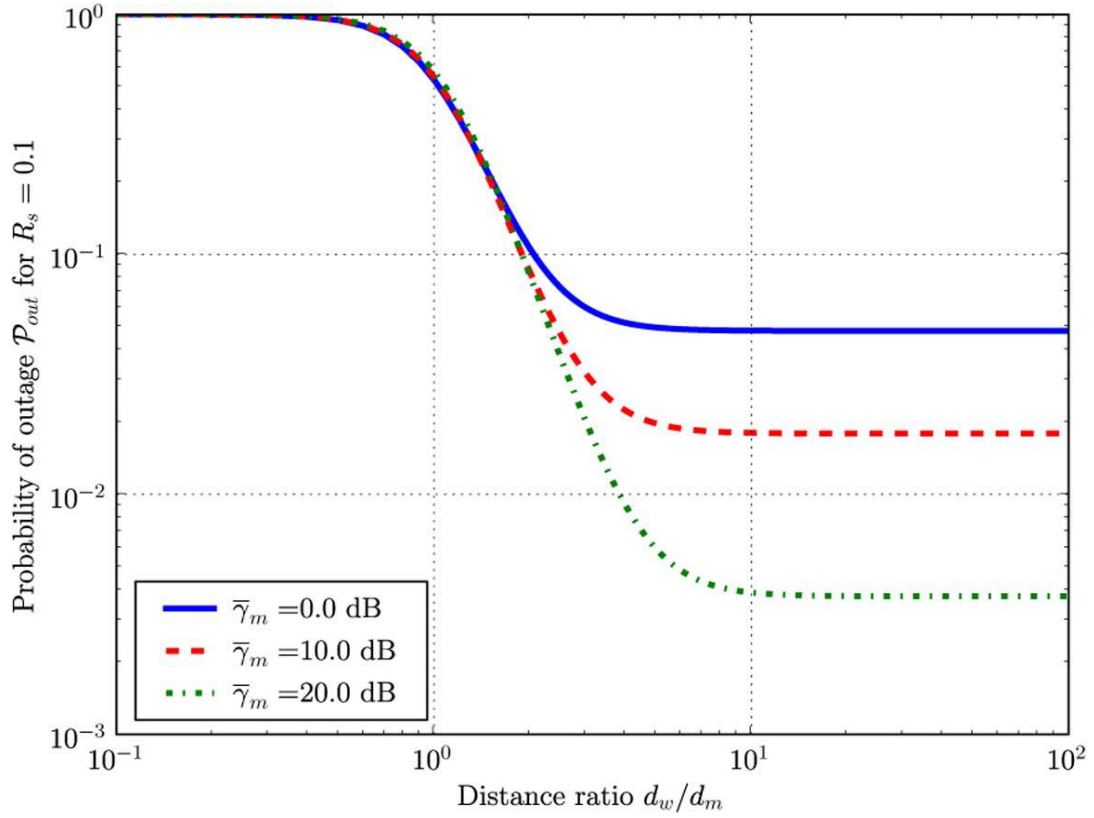


Figure 3.4 Outage probability versus distance ratio

The plot in figure 3.4 indicates outage probability against ratio of distance of eavesdropper channel over main channel distance for specify values of signal to noise ratio of the main channel and standardize target secrecy rate 0.1[23]

The high signal to noise ratio of scheme of outage probability presented as $P_{out}(R_s) \approx (2^{R_s} - 1)/\bar{\gamma}_m$. This indicates probability decays as $\frac{1}{\bar{\gamma}_m}$, otherwise, when $\bar{\gamma}_w \gg \bar{\gamma}_m$ and confidential communication becomes impossible

$$P_{out}(R_s) \approx 1.$$

Figure 3.3 demonstrated the outage probability is low when the signal to noise ration of main channel is large and high when the signal to noise ration of the eavesdropper's channel is large. Conversely, the outage probability decays when signal to noise ration of main channel is greater than that of eavesdropper's channel otherwise it proceed toward one.

Figure 3.4 shows the relationship between outage probability and distance. Outage probability occurs when the transmitter (Alice) make a wrong estimates ($\hat{H}_m$ and $\hat{H}_w$) of the gains of both main channel and eavesdropper's channel. The transmitter (Alice) and the legitimate receiver (Bob) ideal estimates of main channel fading coefficient ($\hat{H}_m = H_m$) is realisable via corporation but eavesdropper (Eve) perhaps will be quiet and transmitter's (Alice) awareness of eavesdropper's (Eve) channel could possibly be rowdy. In an ideal situation eavesdropper's (Eve) fading coefficient by transmitter evaluation is model and presented as

$$\hat{H}_m = H_w + \hat{Z}_w$$

The term H_w is denoted as perfect fading coefficient and term $\hat{Z}_w$ is denoted as zero-mean complex Gaussian noise amidst established variance (σ_e^2) by extent. Outage occurs when the transmitter (Alice) miscalculates gain of the eavesdropper's (Eve) channel in order to realize protected transmission rate which is variance with the channel.

Theorem 3.3 Upper-bounded outage probability is presented as

$$P_{out} \leq \frac{1}{2} - \frac{1}{2} \frac{1}{\sqrt{1 + \sigma_e^2}} \quad (3.10)$$

Proof

[3] Outage Occurs when the transmitter (Alice) miscalculates the extent of confidentiality of message it can refine from opportunistic transmission protocol and presented as

$$P_{out} = P[\hat{C}_s > C_s \geq \Gamma_s, C_m > \Gamma_m] \leq \hat{C}_s > C_s = P[\hat{\Gamma}_w < \Gamma_w]$$

Now, $P[\hat{\Gamma}_w < \Gamma_w]$ can be written as follow

$$P[\hat{\Gamma}_w < \Gamma_w] = \int_0^\infty P(\hat{\Gamma}_w | \Gamma_w) d\hat{\Gamma}_w P(\Gamma_w) d\Gamma_w$$

The term $P(\Gamma_w)$ is denoted as the probability density function of Γ_m and while term $P(\hat{\Gamma}_w | \Gamma_w)$ is denoted as the probability function of $\hat{\Gamma}_w$ constrain by Γ_w . However, these probability density function is expressed as

$$P(\hat{\Gamma}_w | \Gamma_w) = \frac{1}{2\bar{\Gamma}_w \sigma_e^2} - e^{\frac{(\Gamma_w + \hat{\Gamma}_w)}{2\bar{\Gamma}_w \sigma_e^2}} I_0 \left(\frac{\sqrt{\Gamma_w \hat{\Gamma}_w}}{\bar{\Gamma}_w \sigma_e^2} \right), \hat{\Gamma}_w > 0$$

The term $I_0(\cdot)$ denote zeroth-order modified Bessel function of the first kind [26]. The probability density function $P[\hat{\Gamma}_w < \Gamma_w | \Gamma_w)$ is hence scale down to expression presented as

$$P[\hat{\Gamma}_w < \Gamma_w | \Gamma_w = \Gamma_w] = 1 - Q_1(\sqrt{\Gamma_w / \bar{\Gamma}_w \sigma_e^2}, \sqrt{\Gamma_w / \bar{\Gamma}_w \sigma_e^2})$$

The term $Q_1(\cdot, \cdot)$ denote the generalized Marcum Q function [26] and by evaluation of upper bound outage probability via standard integrals is scale down to expression presented as

$$P_{out} \leq \frac{1}{2} - \frac{1}{2} \frac{1}{\sqrt{1 + 2/\sigma_e^2}}$$

Based on this expression the high bound is a reducing function of discrepancy in channel estimation miscalculation (σ_e^2) on outage probability. This show that transmitter (Alice) constantly miscalculate an ideal wiretap (Eavesdropper) fading coefficient and communicate below the normal instant secrecy capacity of the channel. [3]

3.3 Opportunistic Secret key Agreement

I Cisz'ar and J.k'orner outlined secrecy capacity of Gaussian wiretap channel and proved that secure communications above wireless quasi-static fading channels can be realized via codes constructed for Gaussian wiretap channel. But construction of an ideal feasible code for Gaussian wiretap channel is however a challenging phenomenon among researchers[6]. Privacy amplification thus, provide an alternative private key agreement protocol for protected key via common randomness in an easy way[29]. U.maurerr proved and presented that Privacy amplification protocol consist of four step method to secure communications namely, "Randomness distribution, message reconciliation, privacy amplification and protected communication" as illustrated in figure 3.5

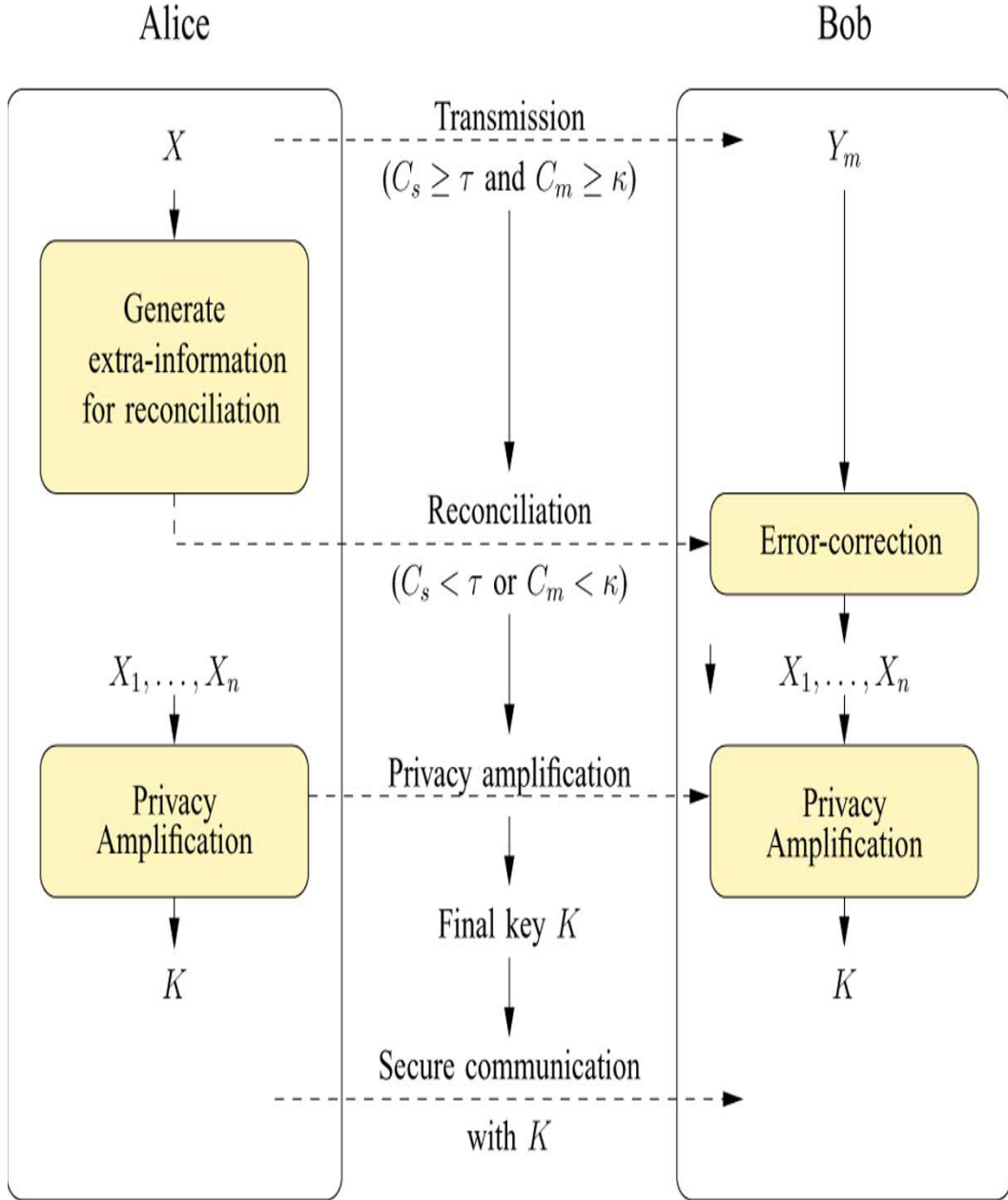


Figure 3.5 Flowchart of the opportunistic protocol [23]

3.3.1 Opportunistic randomness distribution

The concept of this protocol can be describe as follows; the transmitter (Alice) broadcast discrete random latter RVX via wireless channel and legitimate receiver (Bob) and eavesdropper (Eve) detects corresponding latter RV RV Y_m and Y_w respectively. Hypothetically, the level of privacy the transmitter (Alice) and legitimate

receiver (Bob) will be able to extract via common randomness is not equal to zero provided that the legitimate receiver (Bob) did not reveal the content of information to eavesdropper (Eve)[28]. This scheme main objective is provision of one directional private key accord protocol necessitating exchange of information from the transmitter (Alice) to the legitimate receiver (Bob) exclusively. Therefore, secrecy is prerequisite that common randomness of main channel should be greater than common randomness of eavesdropper ($I(X; Y_m) > I(X; Y_w)$). This case indicates a strictly positive secrecy capacity if randomness is shared. The transmitter (Alice) and legitimate receiver (Bob) can strategically utilize variation of instant secrecy capacity (C_s) and share randomness when secrecy capacity (C_s) is large and ideal channel state information of the eavesdropper (Eve) is accessible. Expression below presents how strategic communication is achieved.

$$D(\Gamma) = \{(Y_m, Y_w) : C_s \geq \Gamma, C_m > \mathcal{K}\} \quad (3.11)$$

The term (Γ) make sure the extent of secrecy capacity which can be extracted via randomness while the term ($\mathcal{K}$) make sure link in the mist of transmitter (Alice) and legitimate receiver (Bob).

3.3.2 Key Generation

The concept of key generation are as follows

1. Reconciliation: The transmitter (Alice) broadcast a message and is wrongful observe my legitimate received (Bob) due to difference in their randomness's values, a new information known as error-correction information is exchange to reconcile this variations.
2. Privacy amplification: The transmitter (Alice) and legitimate receiver (Bob) refined privacy bit from reconcile information via privacy amplification scheme[23]

3.3.3 Protected Communication:

The transmitter (Alice) and the legitimate receiver (Bob) utilize this refined privacy key from key generation to exchange information via one way direction in order to guarantee an ideal secrecy as illustrated in figure 11. Perfect estimation of main channel and eavesdropper channel evaluation of instant secrecy capacity and extent of privacy key to distill significantly, depends on this two way of key generation (randomness distribution and privacy amplification) [23]

3.4 Review of Secret-key accord for practical scheme

Secret-key agreement protocol for an ideal Gaussian wiretap channel for secure communication procedure is as follows.

3.4.1 Protected communication procedures

Protected communication procedures are enhanced by an accord between the transmitter (Alice) and legitimate receiver (Bob). The transmitter (Alice) generate an allocate randomness in form of transmitting sequence represented as

$X^n = (X_1 \dots X_n)$ where n are equally distributed attainment of a discrete random variable X across the main channel and does not depends on each other and equally distributed. Both the legitimate receiver (Bob) and Eavesdropper (Eve) observes a correlated sequence of continuous random variables as resulted of this attainment represented as

$$Y_m^n = (Y_{m,1} \dots Y_{m,n}) \text{ and } Y_w^n = (Y_{w,1} \dots Y_{w,n})$$

Estimation Problem.

Both The transmitter (Alice) and legitimate receiver (Bob) exhibit errors in estimation as a result of rowdiness of the channel presented

X^n Transmitter symbol (Alice), and legitimate receiver's symbol (Bob)
 $X'^n = (\hat{X}_1, \dots, \hat{X}_n)$

Reconciliation

The transmitter (Alice) and legitimate receiver (Bob) first attempt is to rectify their wrong estimation by transmitting a new reconcile message under the high premise that this information is completely accessible to the eavesdropper (Eve). This reconciliation is presumed that transmitter (Alice) reduce its source latter X^n and legitimate receiver (Bob) decrypted via correlated side information Y_m^n .

Small bound on the overall bits M_{rec} that should be exchanged between the transmitter (Alice) and legitimate Receiver (Bob) given by Slepian –Wolf theorem and presented as [30]

$$M_{rec} \geq H(X^n|Y_m^n) = nH(X|Y_m) \quad (3.12)$$

An overhead $\epsilon_{rec} > 0$ is initiated by practical reconciliation scheme where the transmission or reconciliation are outlined and presented as

$$M_{rec} = nH(X|Y_m)(1 + \epsilon_{rec})$$

$$\beta = 1 - \epsilon_{rec} \frac{H(X|Y_m)}{I(X; Y_m)} \leq 1 \quad (3.13)$$

Perquisites bits rate for reconciliation is presented as

$$M_{rec} = n(H(X) - \beta I(X; Y_m)) \quad (3.14)$$

the transmitter (Alice) and legitimate receiver (Bob) after the reconciliation allocate eminent probability of a common sequence with entropy represented as X^n and $n_{rec} = nH(X)$. This common sequence is reduced to a binary sequence represented as S which has a length n_{rec} [31]. Two protocols are employ for reconciliation and error rectification namely multilevel coding and Multistage Decoding.

Amplification

Confidentiality amplification allows the transmitter (Alice) and legitimate receiver (Bob) to decode a privacy key from binary sequence (S) [32]

Problem Statement

The transmitter (Alice) should chose X achieving the capacity $y \ 0.5 \log_2(1 + Y_m)$ of the main channel. This can only be achieve via optimization in order for common randomness to approaches the channel capacity.

Chapter 4. METHODOLOGY AND DISCUSSION OF RESULTS

4.1 Introduction

This chapter builds upon analysis and review of fundamental of information theoretic security scheme highlighted in chapter three. Closer form outline of secrecy capacity (SC) performance metric for security in wireless communication networks is characterized via monte-Carlo simulation for different scenarios. This problem is analysed by considering a three-terminal communication scenario as shown below.

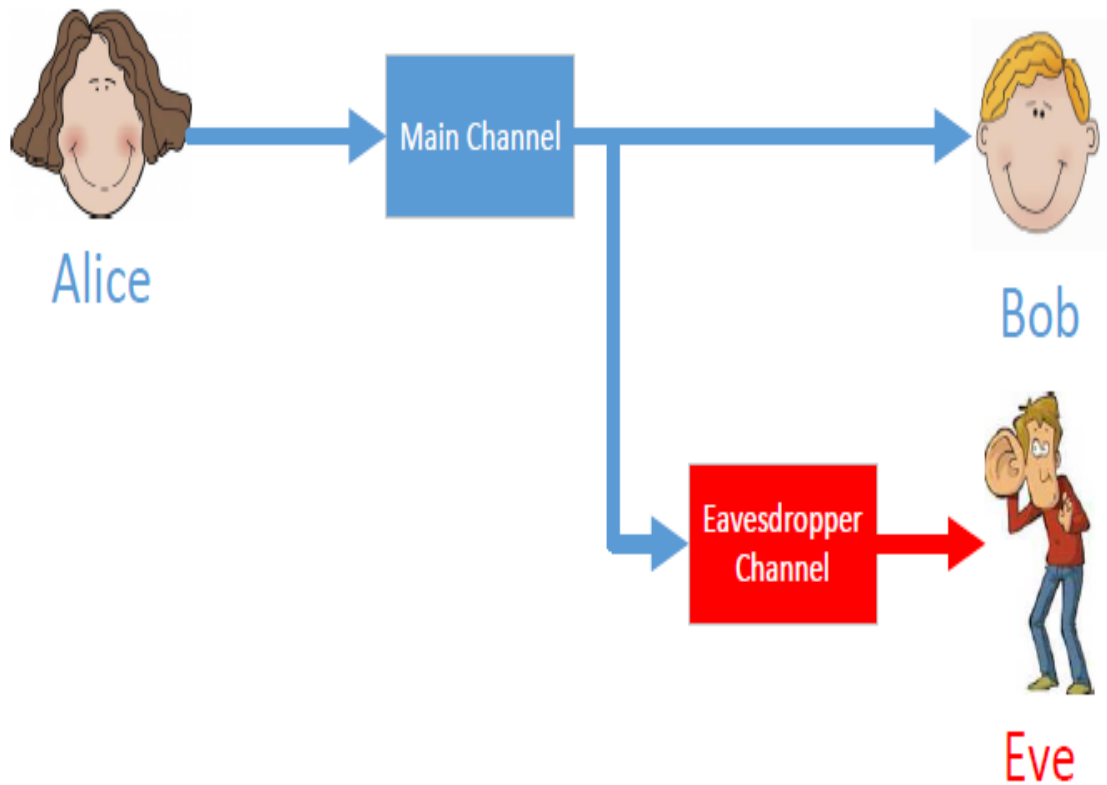


Figure 4.1 Simple Wiretap channel [36]

Problem statement

The transmitter (Alice) intends to broadcast information to a legitimate receiver (Bob) via the main channel which is wiretap by an eavesdropper (Eve) via a deteriorate version called eavesdropper channel. The question then arise in two way; firstly the extent of rate in which the transmitter (Alice) could communicated to the legitimate

receiver (Bob) and secondly the extent of rate in which eavesdropper (Eve) could wiretap on Alice communication (Eavesdropper channel RAE).

However, with these knowledge of observation by channel capacity, specified secrecy capacity is calculated based on variation of rates of main and eavesdropper's channel. Thus, strong security scheme is required to protect against this vulnerabilities [36]

Therefore, physical layer security provides an alternative means to strength the traditional security scheme (cryptography) by exploiting the channel physical features to secure communication.

4.2 Systems architecture for a simply wireless sensor network

The systems architecture for a wireless sensor network as illustrated in figure 4.1

This consist of three node. The transmitter (Alice) intends to broadcast an information to a legitimate receiver (Bob) where malicious eavesdropper (Eve) is privy of their exchange of information. The term 'a' denote the transmitter (Alice), 'b' denote legitimate receiver (Bob) and 'e' denote eavesdropper (Eve). The term x_j^a , y_j^b and z_j^e denotes the transmitter latter for the transmitter (Alice), legitimate receiver(bob) and eavesdropper (Eve). A premised is made between the transmitter (Alice) and both legitimate (Receiver) and eavesdropper (Eve) to be the same Rayleigh fading.

The term H_{xy} is a complex zero-mean Gaussian random variable denoted as the modified $E[|H_{xy}|^2] = 1$ multipath fading. The premise is made that during entire during of code multipath fading is constant and independent from both transmitter (Alice), legitimate receiver (bob) and eavesdropper (Eve) pair. The signal observed by both legitimate receiver (Bob) and eavesdropper (Eve) is presented as

$$y_j^b = \frac{H_{a,b}}{\sqrt{d_{a,b}^\alpha}} \sqrt{E_s} x_j^a + n_j^b$$

$$y_j^e = \frac{H_{a,e}}{\sqrt{d_{a,e}^\alpha}} \sqrt{E_s} x_j^a + n_j^e$$

The term $d_{a,b}$ denotes the distance between the transmitter (Alice) and legitimate receiver (Bob), $d_{a,e}$ denotes the distance between the transmitter (Alice) and eavesdropper (Eve), α denotes path-loss exponent, E_s denotes transmitted energy per latter n_j^b and n_j^e , $H_{a,b}$ denotes the channel gain of the transmitter(Alice) and legitimate receiver(Bob) , and $H_{a,e}$ denotes the channel gain of the transmitter (Alice) and eavesdropper(Eve).When multiple packets are considered in the ARQ systems ,it will be assumed that the fading affecting different packets is independent[36]

4.3 Simulation Results and Evaluation of the model

This section start by numerically analysing the lower and upper bound of secrecy capacity as the transmitter and receiver are fixed in a passion while the eavesdropper moving round. In this report, four scenarios are consider in whose parameter are described in in tables below .The first set represents a vulnerability architecture for which a compromised access point is likely to use to launch an attack when eavesdropper (Eve) very close to the transmitter (Alice).The second and third set of parameter describe a more robust, where secrecy capacity depends on path-loss exponent and distance ratio. The secrecy capacities of many networks assignment in these case scenarios are analyses are shown in table I to VI. All results are obtained under the following assumption.

1. The transmitter is in a fixed point
2. Receiver is fixed in a position
3. No fading and randomness is consider
4. Location of the malicious eavesdropping determine secrecy capacity intensity.
5. Secrecy capacity level (High or low) is determined at each location of eavesdropper.

Results

Scenario A

Case I show the impact of secrecy capacities is zero when the eavesdropper (Eve) is at the same coordinate with legitimate receiver (Bob) for different wireless path loss-exponents as illustrated in table 1 and figure 4.2. This signifies vulnerability region and security is compromised

Case II shows the intensities of the secrecy capacities increase as the eavesdropper at coordinate (1, 5, 1, 5) is moving away from the legitimate receiver at a fixed coordinate (0.5, 0.015) for the different path-loss exponent values as indicated in the table II and figure 4.3 This security is guarantee at this region.

Case III Demonstrated extent of secrecy capacities further increase as the eavesdropper(Eve) vanish at coordinate(1.7,17) and legitimate receiver(Bob) at a fixed coordinate(0.5,0.015) for different path-loss exponents as illustrated in table III and figure4.4. This once again demonstrate the impact of distance of user location and path-loss exponent on secrecy capacity

Scenario B

Case I Shows the increment on the secrecy capacities as the eavesdropper is fixed coordinate (1.5, 1.5) and the legitimate receiver coordinate changes at (0.5, 0.025) as illustrated in Table IV and Figure 4.5. For different values of path-loss exponent. This demonstrate the impact of distance and path-loss on secrecy capacities

Case II Show the secrecy capacities intensities as the eavesdropper move further away at coordinates (1.6,1.6) while the legitimate receiver at fixed coordinate (0.5,0.035) as illustrated in table V and figure 4.6. For different path loss-exponent.

Case III Show further show the secrecy capacities increment and the eavesdropper vanish at coordinate (1.7, 1.7) and the legitimate receiver is at fixed coordinates (0.5, 0.055) as illustrated in table VI and figure 4.7. Once again demonstrate the impact of position and path-loss exponent on secrecy capacities.

As indicated in table I to VI Secrecy capacity defines the tightest upper bound on the amount of information that can be reliably and securely transmitted over a communication channel enhance the understanding of performance limits of the wire-tap channel systems and thus serve as an instruction guideline for the performance optimization and engineering of such systems.

SCENARIO A

Case 1 **Table I**

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.015	0.5, 0.015	2.5	0
0, 0	0.1, 0.1	0.5, 0.015	0.5, 0.015	2.7	0
0, 0	0.1, 0.1	0.5, 0.015	0.5, 0.015	3.5	0
0, 0	0.1, 0.1	0.5, 0.015	0.5, 0.015	4.5	0
0, 0	0.1, 0.1	0.5, 0.015	0.5, 0.015	5.5	0

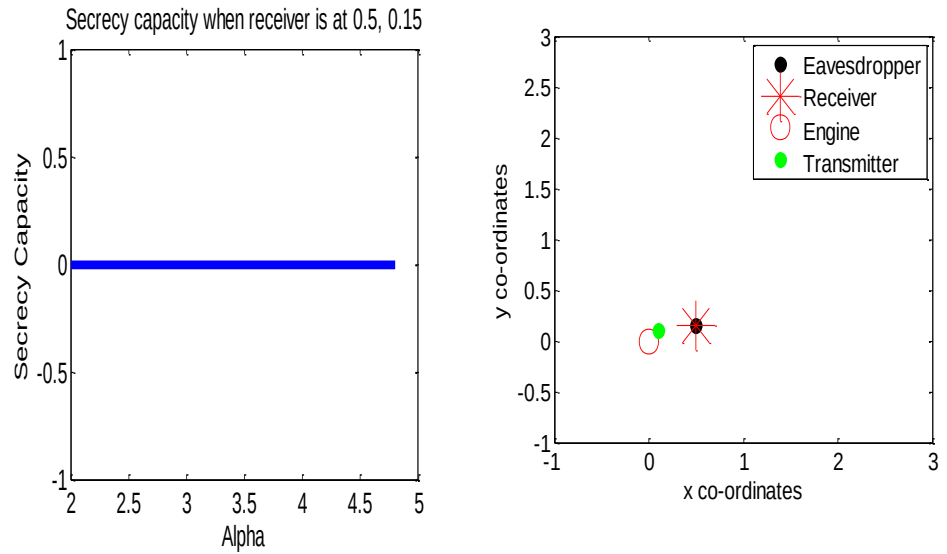


Figure 4.2

Case 2 Table II

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.015	1.5, 1.5	2.5	2.4041
0, 0	0.1, 0.1	0.5, 0.015	1.5, 1.5	2.7	2.5964
0, 0	0.1, 0.1	0.5, 0.015	1.5, 1.5	3.5	3.3657
0, 0	0.1, 0.1	0.5, 0.015	1.5, 1.5	4.5	4.3273
0, 0	0.1, 0.1	0.5, 0.015	1.5, 1.5	5.5	5.2890

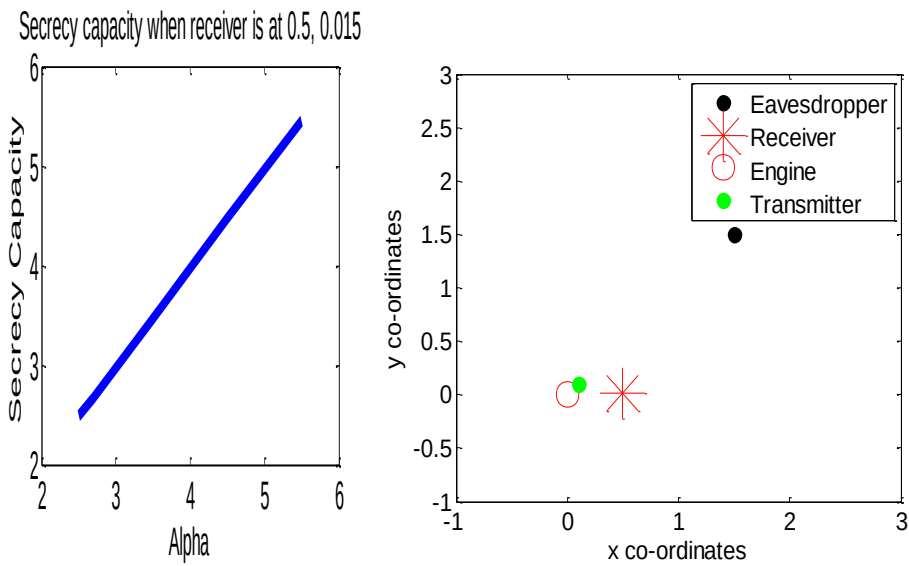


Figure 4.3

Case 3 Table III

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.0 15	1.6, 1.6	2.5	2.5285
0, 0	0.1, 0.1	0.5, 0.0 15	1.6, 1.6	2.7	2.7308
0, 0	0.1, 0.1	0.5, 0.0 15	1.6, 1.6	3.5	3.5399
0, 0	0.1, 0.1	0.5, 0.0 15	1.6, 1.6	4.5	4.5513
0, 0	0.1, 0.1	0.5, 0.0 15	1.6, 1.6	5.5	4.5627

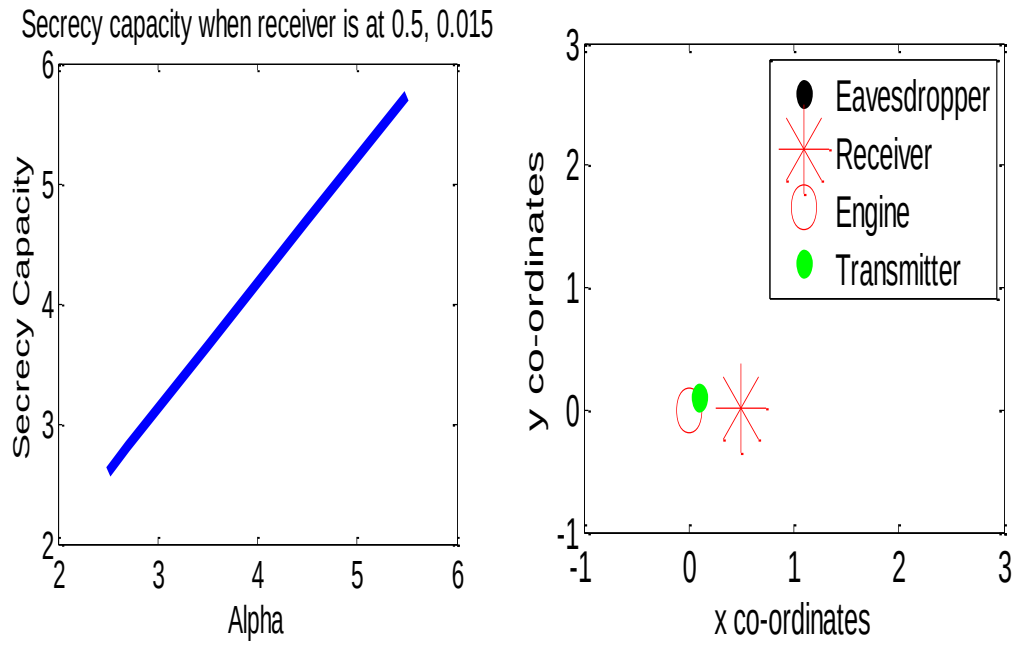


Figure 4.4

SCENARIO B

Case 1 **Table IV**

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.025	1.5, 1.5	2.5	2.4795
0, 0	0.1, 0.1	0.5, 0.025	1.5, 1.5	2.7	2.6779
0, 0	0.1, 0.1	0.5, 0.025	1.5, 1.5	3.5	3.4713
0, 0	0.1, 0.1	0.5, 0.025	1.5, 1.5	4.5	4.4632
0, 0	0.1, 0.1	0.5, 0.025	1.5, 1.5	5.5	5.4550

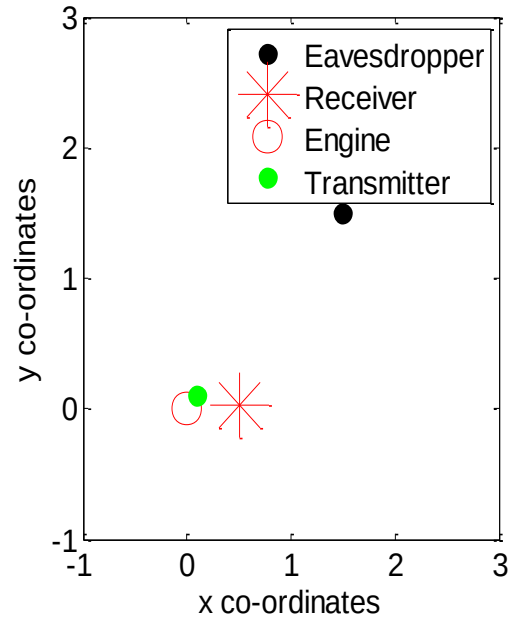
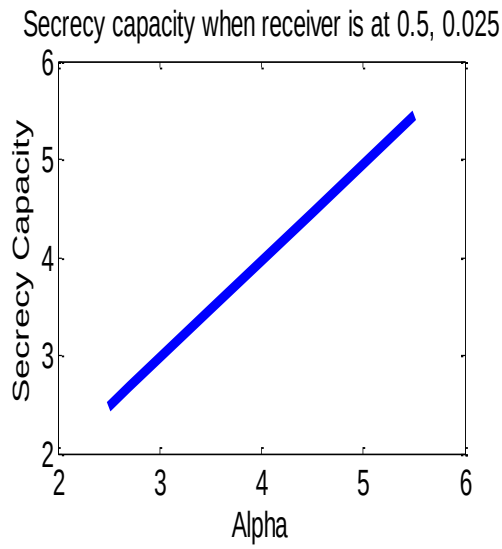


Figure 4.5

Case 2 Table V

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.035	1.6, 1.6	2.5	2.6018
0, 0	0.1, 0.1	0.5, 0.035	1.6, 1.6	2.7	2.8099
0, 0	0.1, 0.1	0.5, 0.035	1.6, 1.6	3.5	3.6425
0, 0	0.1, 0.1	0.5, 0.035	1.6, 1.6	4.5	4.6832
0, 0	0.1, 0.1	0.5, 0.035	1.6, 1.6	5.5	5.7240

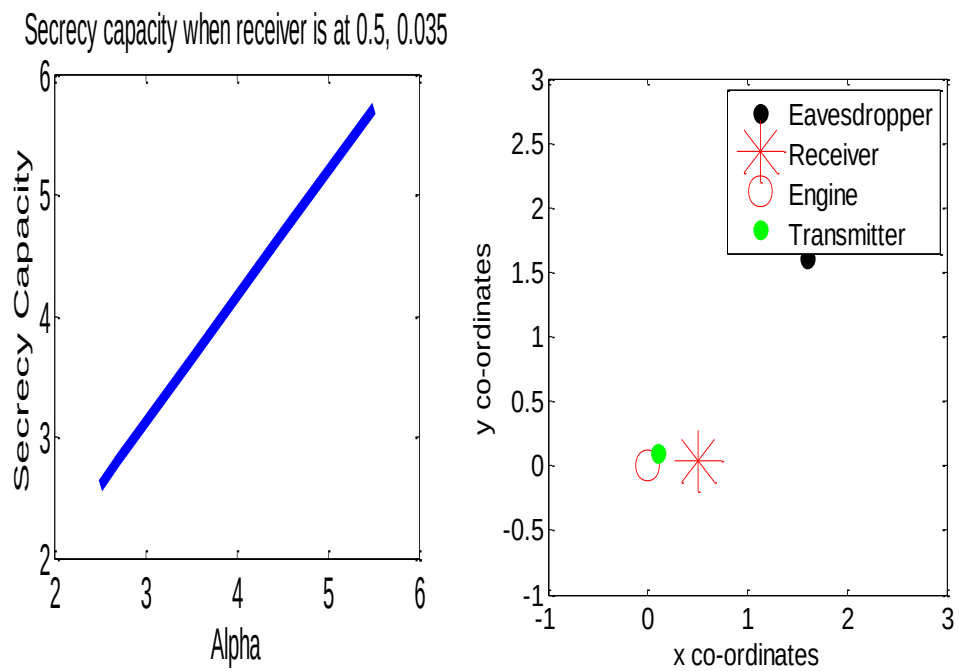


Figure 4.6

Case 3 Table VI

Engine Coordinates	Transmitter Coordinates	Receiver Coordinates	Eavesdropper Coordinates	Alpha	Secrecy Capacities
0 , 0	0.1, 0.1	0.5, 0.055	1.7, 1.7	2.5	2.7117
0, 0	0.1, 0.1	0.5, 0.055	1.7, 1.7	2.7	2.9287
0, 0	0.1, 0.1	0.5, 0.055	1.7, 1.7	3.5	3.7964
0, 0	0.1, 0.1	0.5, 0.055	1.7, 1.7	4.5	4.8811
0, 0	0.1, 0.1	0.5, 0.055	1.7, 1.7	5.5	5.9658

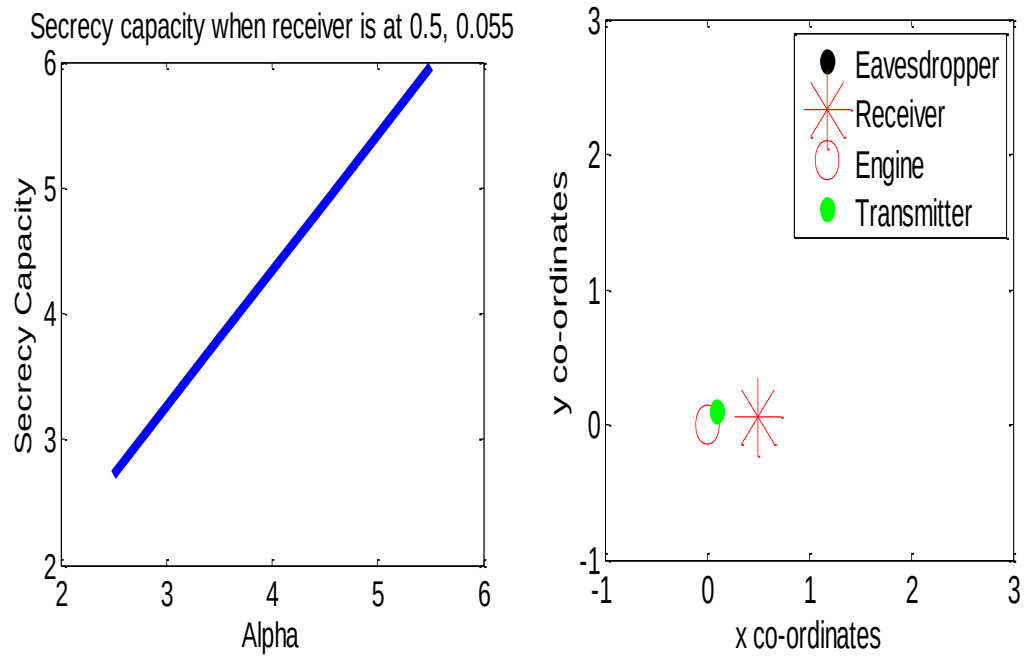
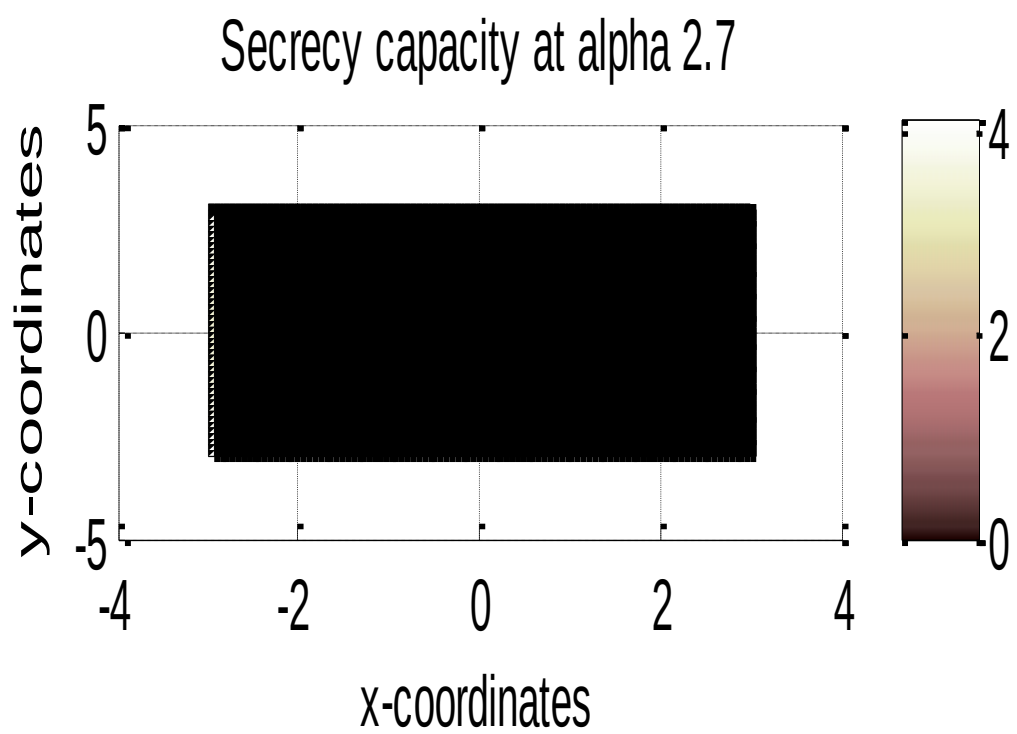
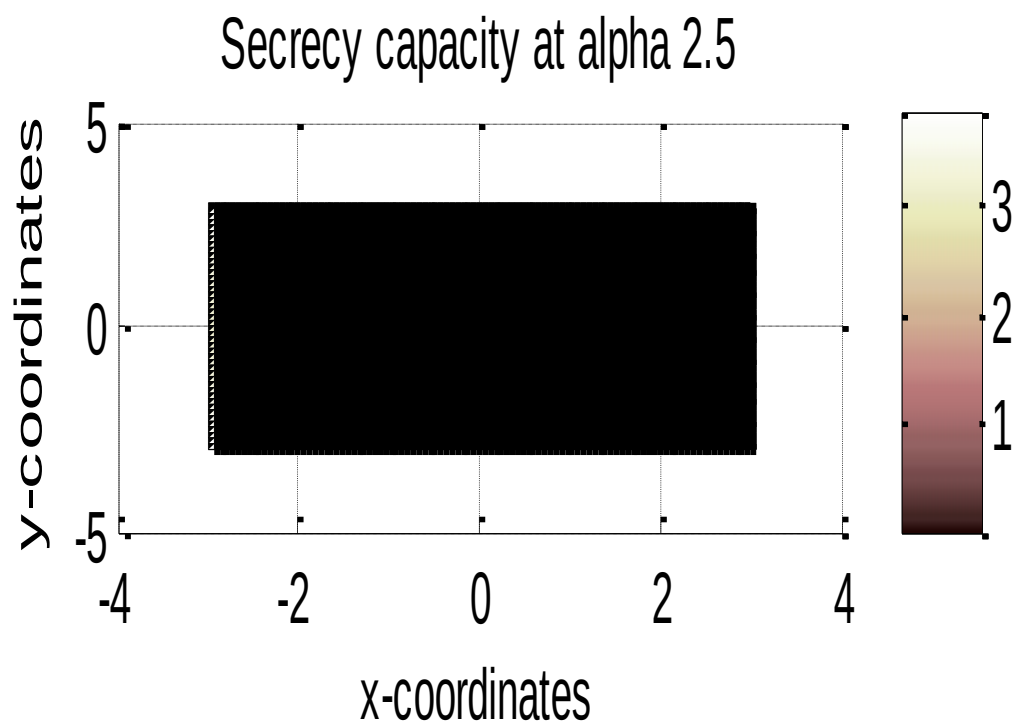
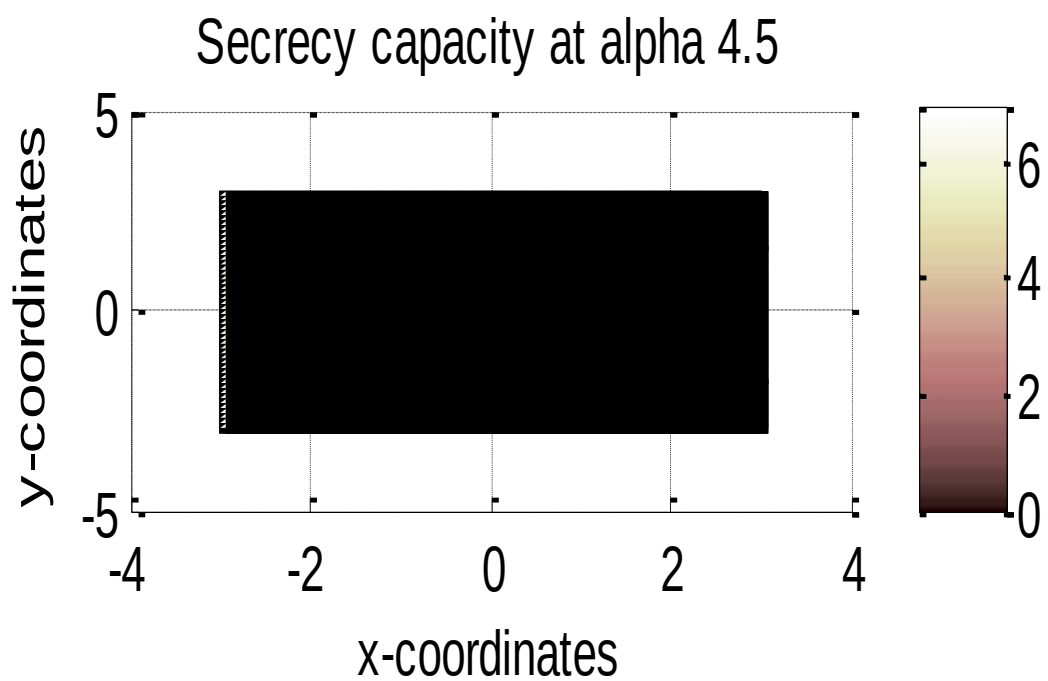
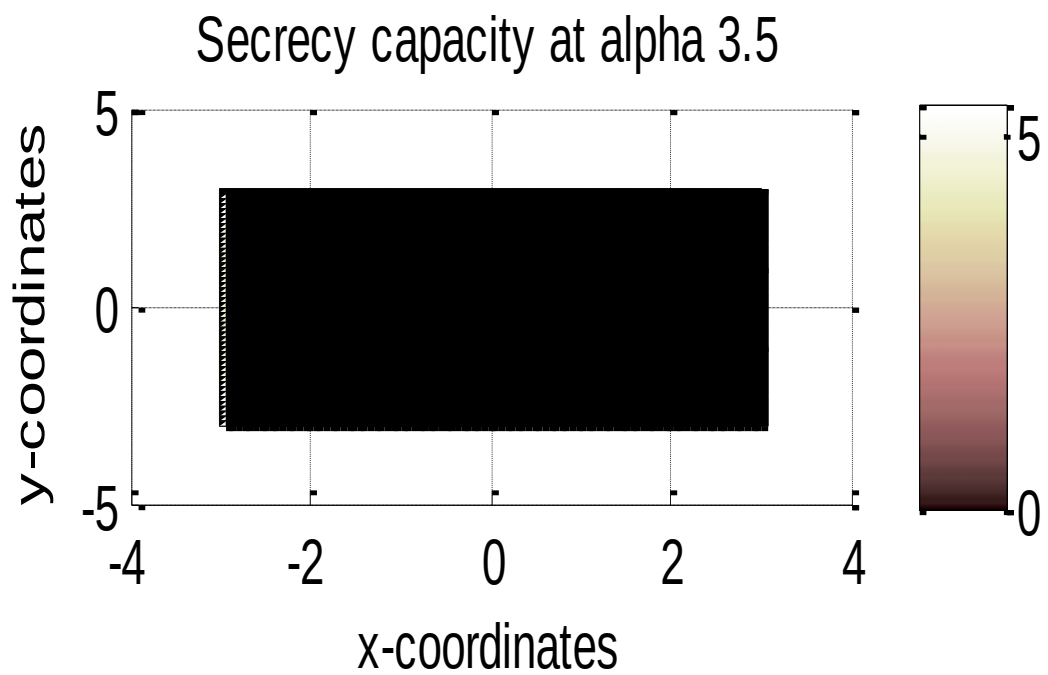


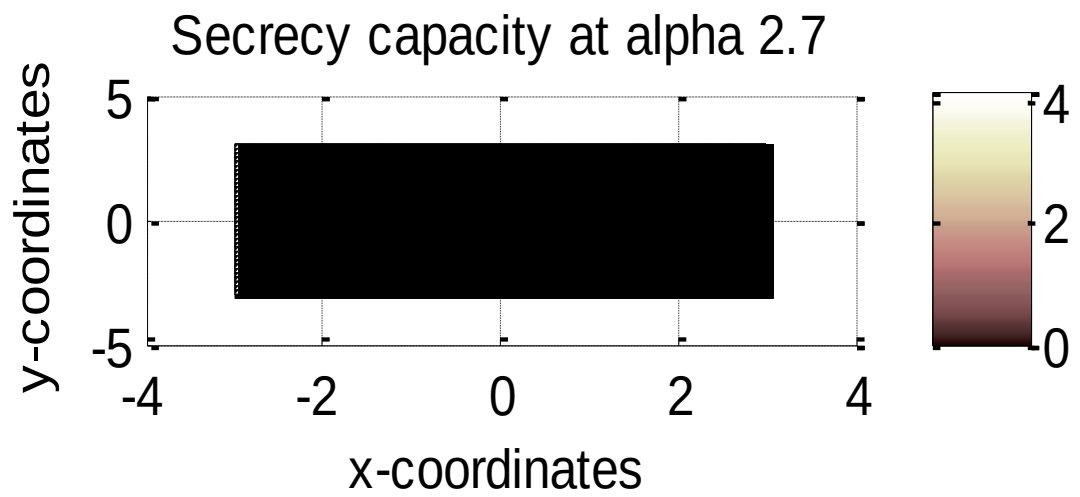
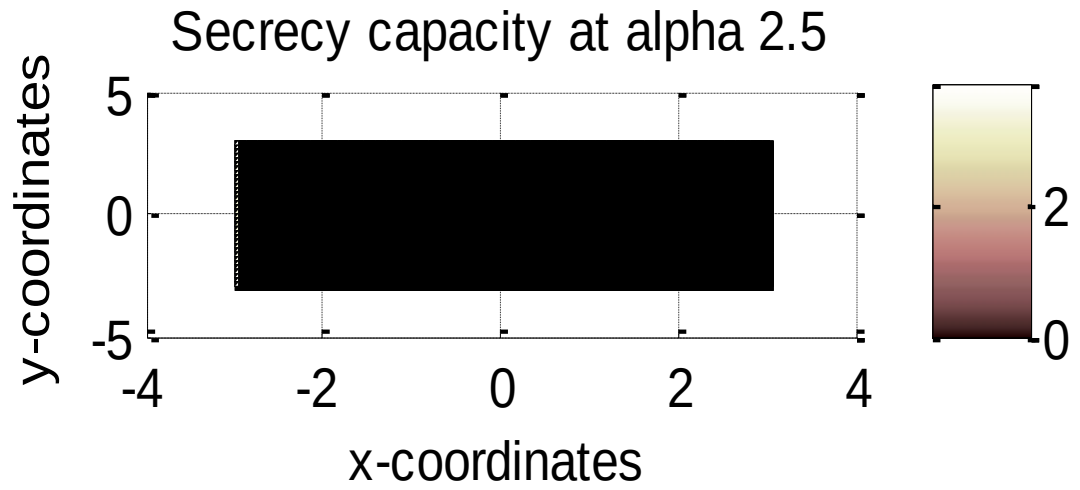
Figure 4.7

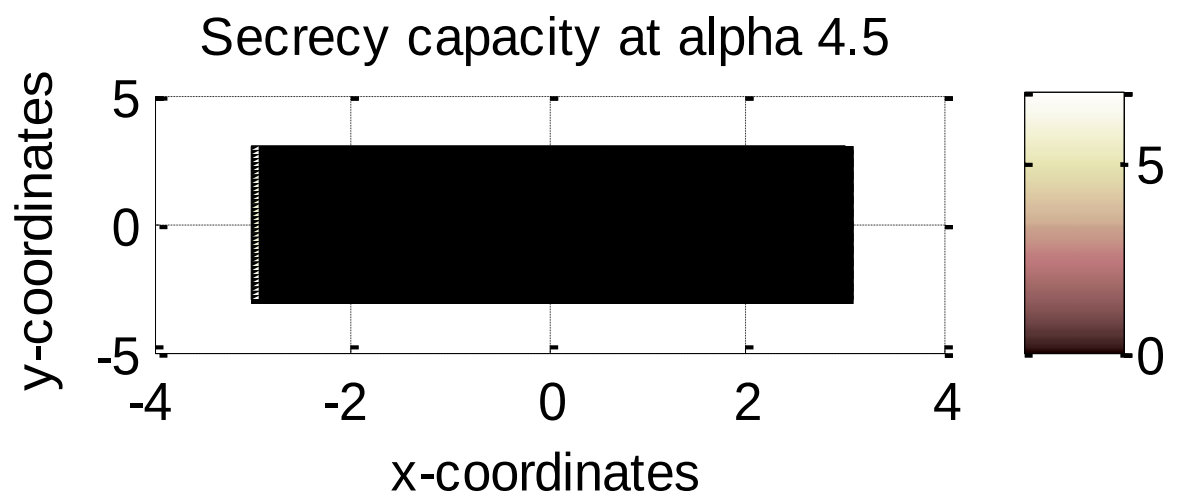
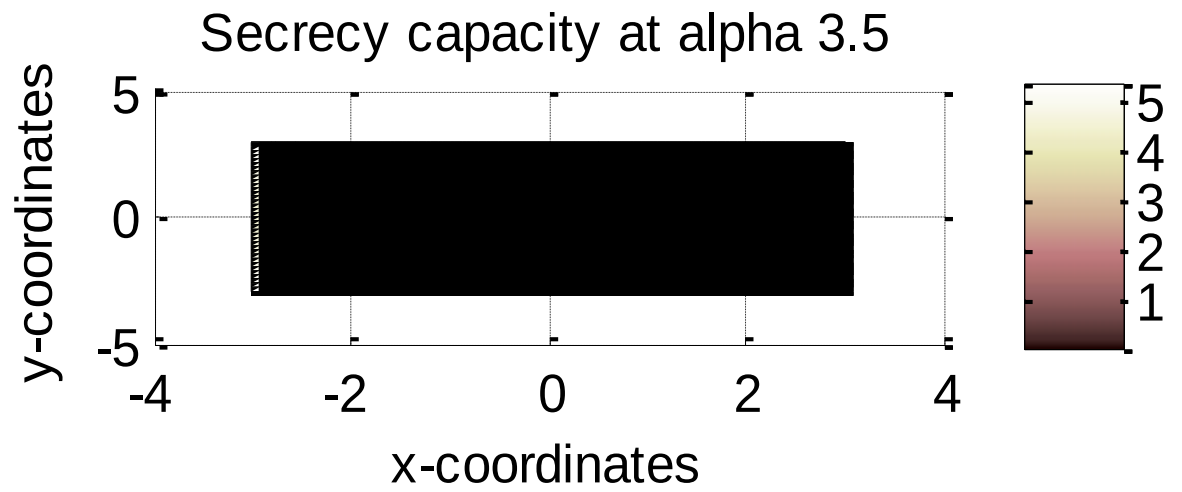
3D plot at legitimate receiver coordinates (0.5, 0.015)



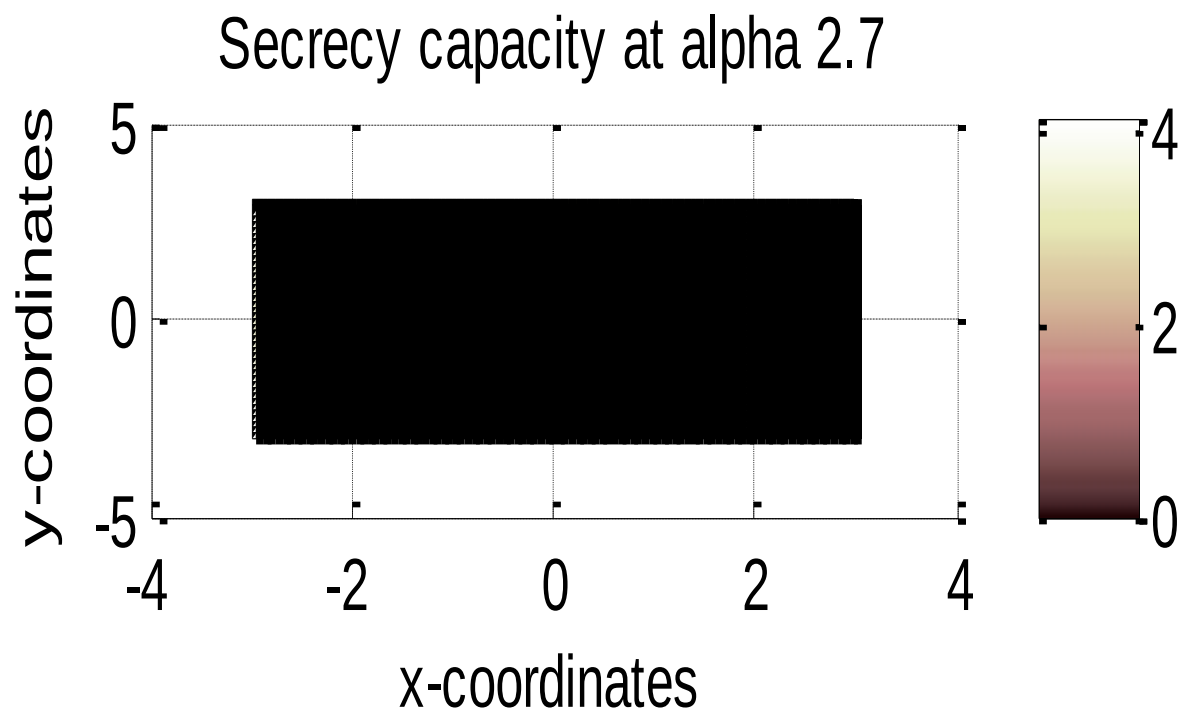
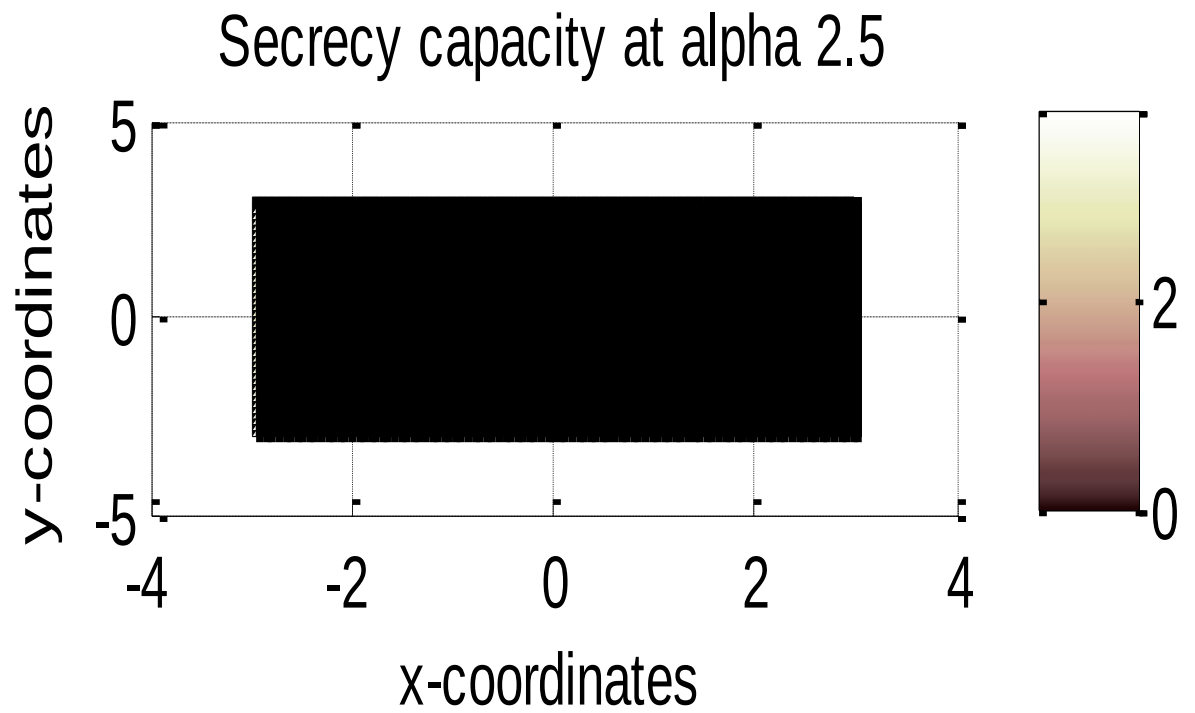


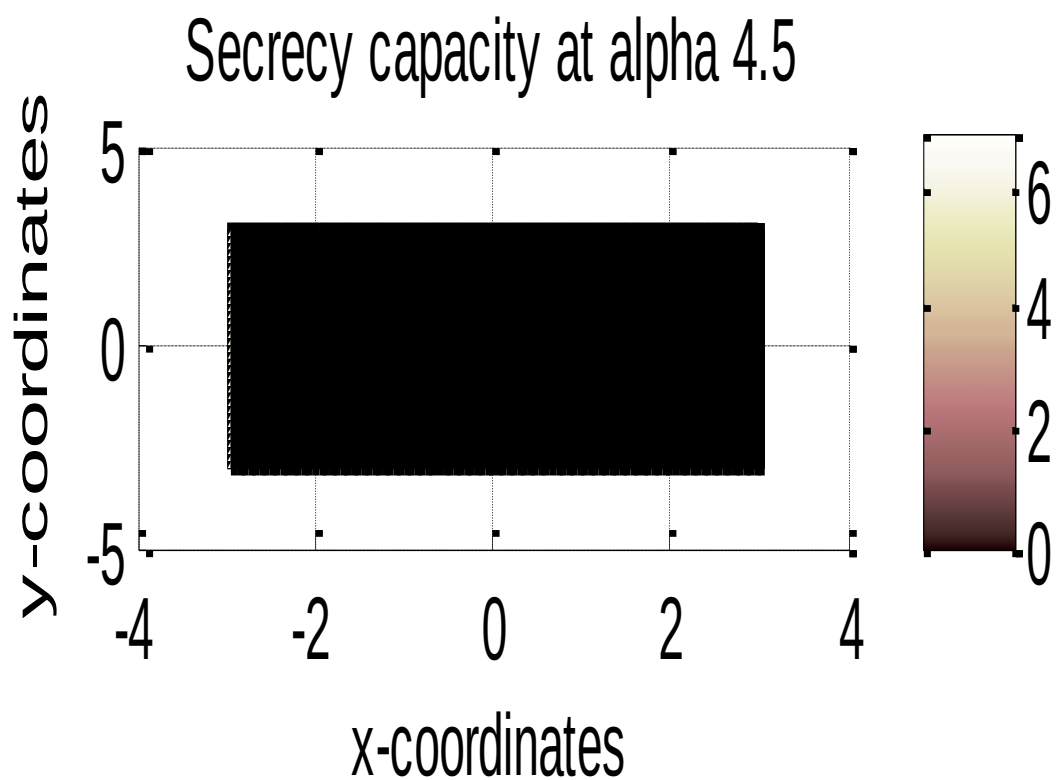
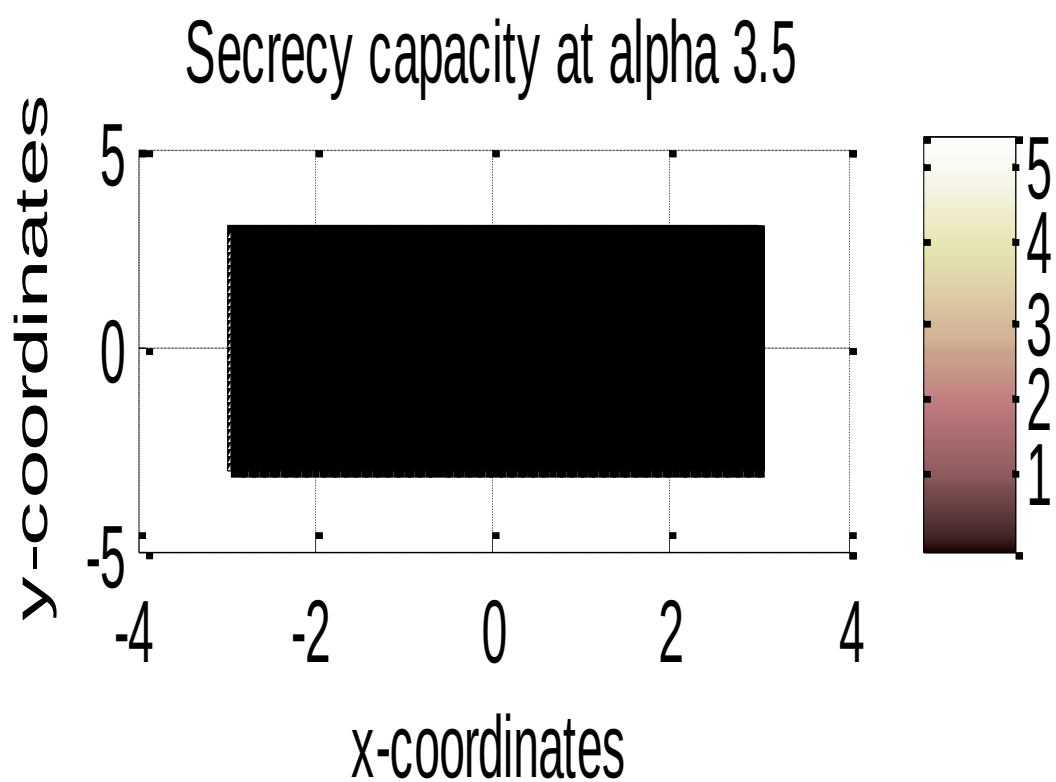
3 D at legitimate receiver coordinates (0.5, 0.25)



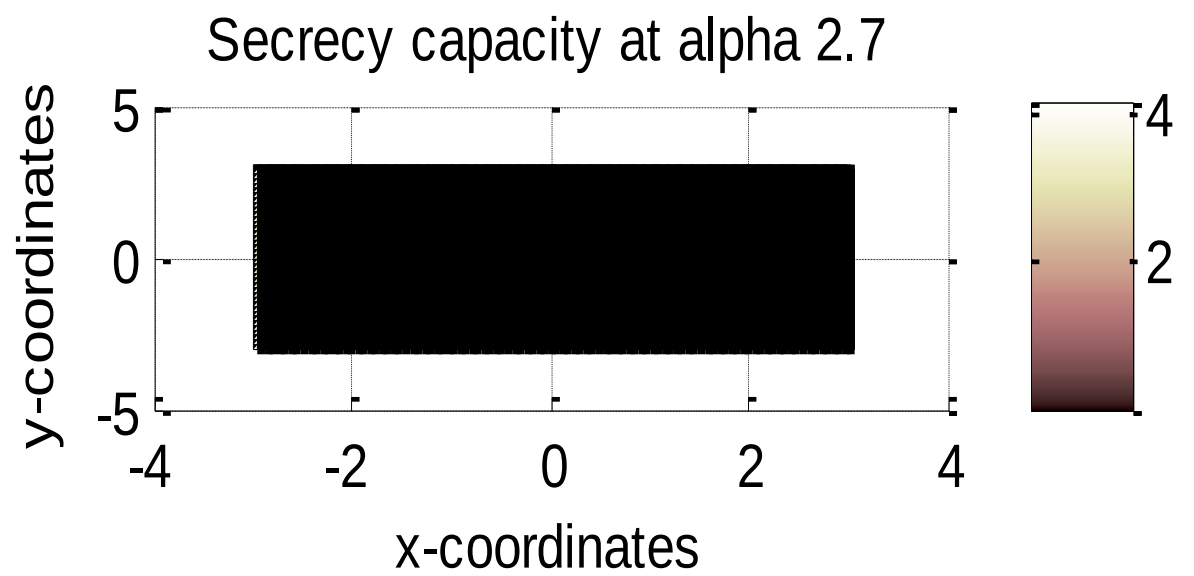
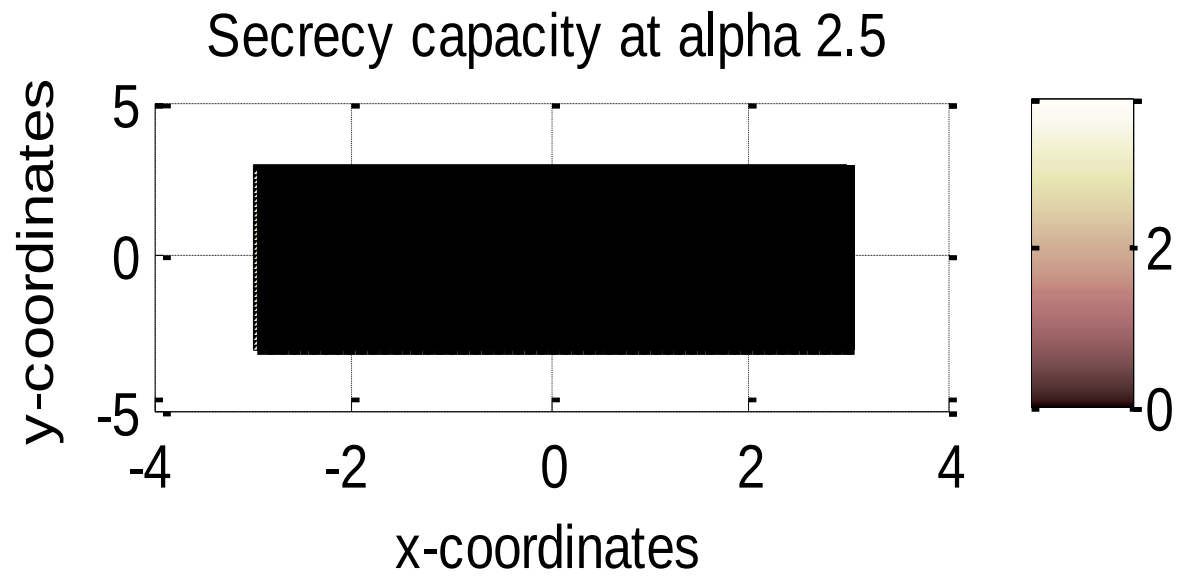


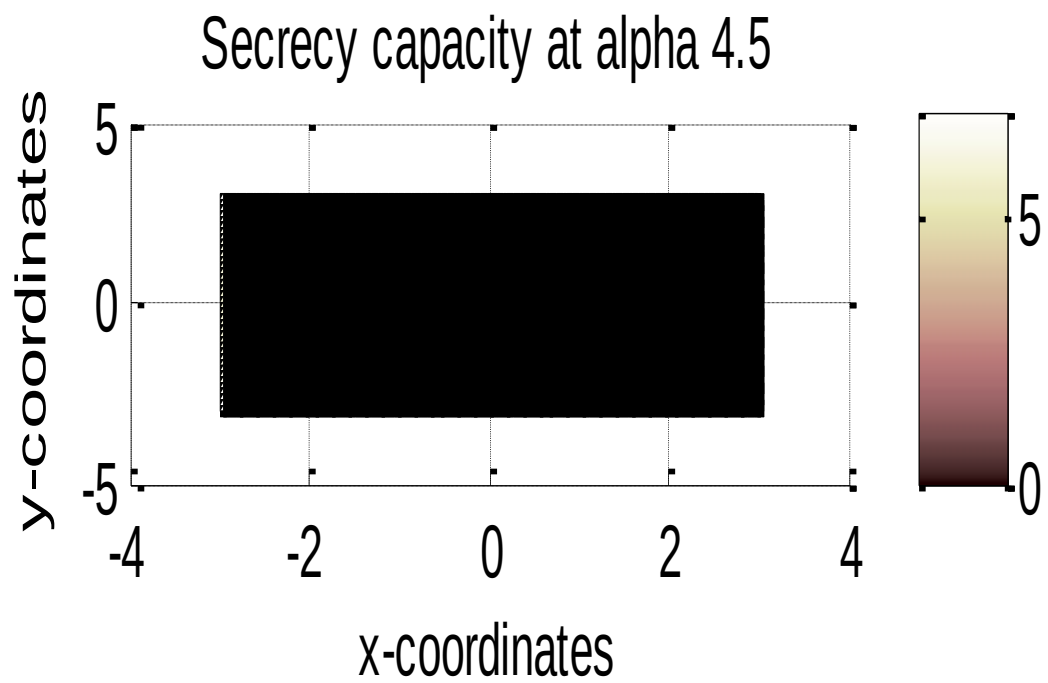
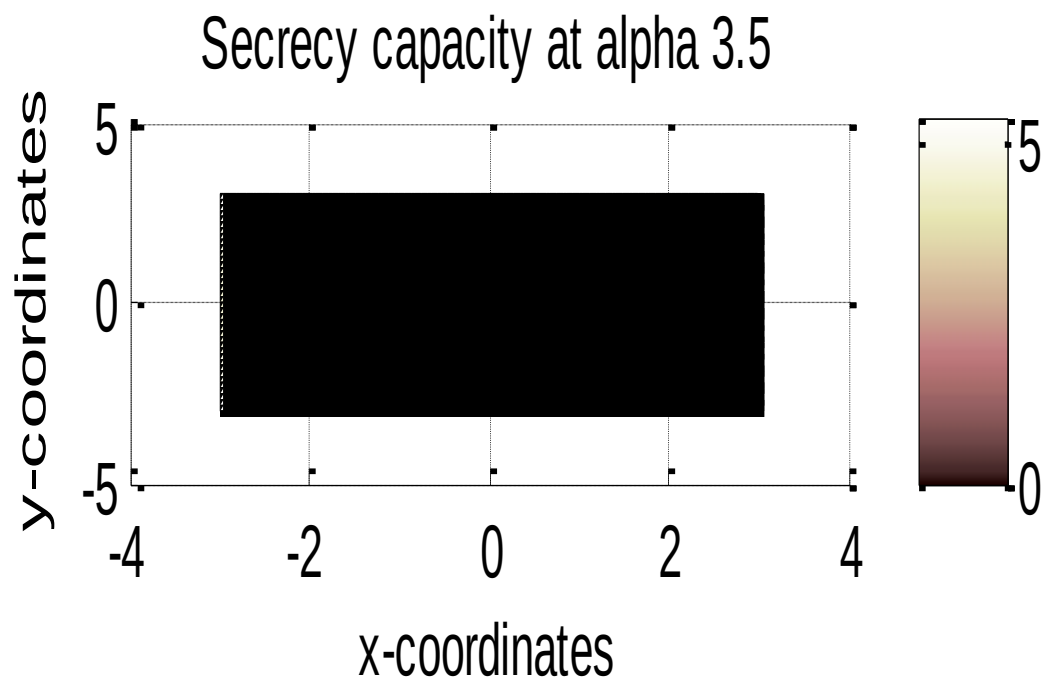
3D at legitimate receiver coordinates (0.5, 0.35)





3D at legitimate receiver coordinates (0.5, 0.55)





CHAPTER 5 BENEFITS AND LIMITATION OF CRYPTOGRAPH AND INFORMATION –THEORETIC SECURITY IN WIRELESS SENSOR NETWORK

This chapter review significances and limitation of cryptography and information theoretic security in different wireless platform.

Table VII classical Cryptography benefits and limitations

Benefits	Limitation
1. Public-key systems like RSA (cryptosystem) are on premise to be protect in a large scale application because to date no record of an attack has occur to compromise this scheme[23] 2. The level of protection is realized via block to block basis [23] 3. Authentication is realizable via Public key cryptography[23] 4. Cryptography scheme are easily accessible and cheap[23]	These scheme based on premise that has not be proven. Hence, security of systems is completely vulnerable to sophisticated attacks and not protected if the premise are wrong[23] This scheme has a short come in term of reliability and security as a platform for block length and cryphertext messages. Confidentiality of cryptographic scheme depends on how it can sustain the level of attack The communication of both transmitter (Alice), legitimate receiver (Bob) and eavesdropper (Eve) is rowdy due to zero secrecy capacity of application layer. Hence, basic cipher is generally devoid of information theoretic security.

	The cryptography scheme rely heavily on sophisticated computation which not feasible in present day wireless communication systems due to advance in technology.
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Table VIII Information Theoretic Security benefits and limitations

Benefits	Limitation
1. The eavesdroppers where not put under any restriction on movement[23] 2. Actual communication wiretapped about where about of an attacker (Eve) can be established[38] 3. This scheme is fully implemented via quantum key distribution[39] 4. Privacy amplification is realized in theory which could be nearer to perfect secrecy[38]	The scheme based on premise of probability which can easily be manipulated for some degree of protection[23] This rely heavily on hypothesis that may not be realistic in practice[23] The technology of this scheme is not easily accessible and very costly.

This short differences in cryptography and information-theoretic security provide an insight in which implementation of physical-layer security could complement and enhance the security of digital communication systems that is yet to be explored.[23]

Chapter 6. CONCLUSION AND REVOMMENDATIONS

The exchange of confidential information between the transmitter (Alice) and legitimate receiver (Bob) via quasi-static fading channel using information theoretic security as problem setup and wiretap of communication by eavesdropper (Eve) via a degraded version of quasi-static fading channel were reviewed and understood in chapter 3. Two performance metric that are mostly use in information theoretic security average secrecy capacity and outage secrecy capacity were specifically reviewed .Also reviewed is four step secure communication protocol.

In view of this fundamentals a simple aspect of physical –layer security is investigated in this dissertation with focus on implementation of practical schemes. The objectives of this dissertation focuses on wireless channels whose specific nature is ideally suited for design of effective physical-layer security for engine monitoring environment.

In summary, a simple wiretap channel model was considered and shown that secrecy capacity which is performance measure for security in information-theoretic security depends on the path-loss and distance ratio, when the user location is of the interest. Fixed coordinates for transmitter (Alice) and legitimate receiver (Bob) were used, while different co-ordinates for the eavesdroppers (Eve) were used which demonstrate that security of wireless sensor network can be complemented and enhanced if the eavesdropper (Eve) is kept some distance away from the transmitter and receiver. Additionally, benefits and limitations of both cryptography and information technology were highlighted in chapter 5, thus implementation of physical layer security could complement and enhanced security of wireless sensor network.

An investigation to conveniently find the optimal rate pairs on both transmitter and receiver links for different intercept probability constraint to effect the maximum end-to end throughput binary search algorithm is recommended for future work.[36]

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